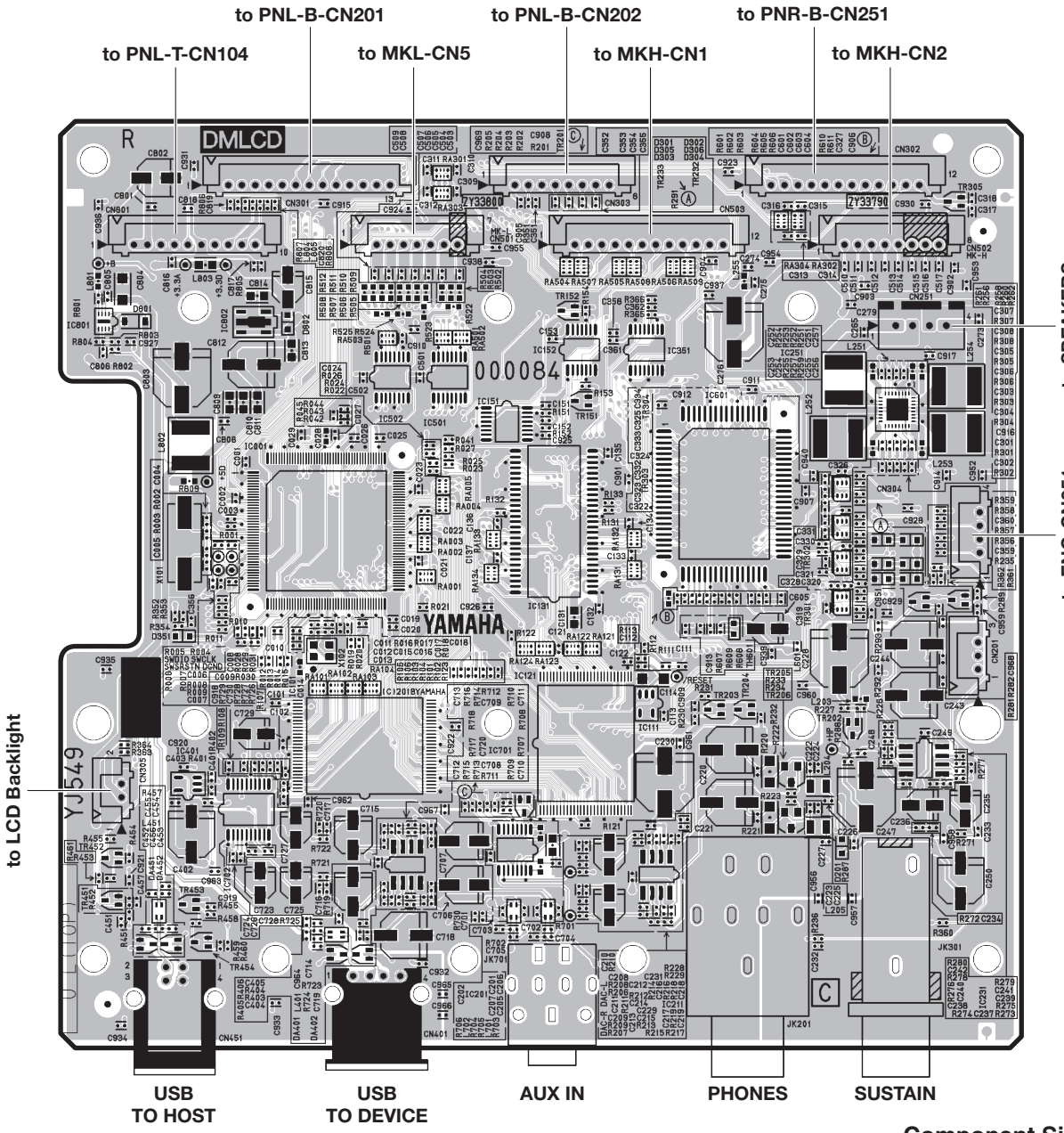
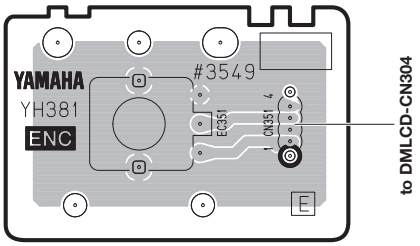
## • DMLCD Circuit Board




**Component Side**（部品側）

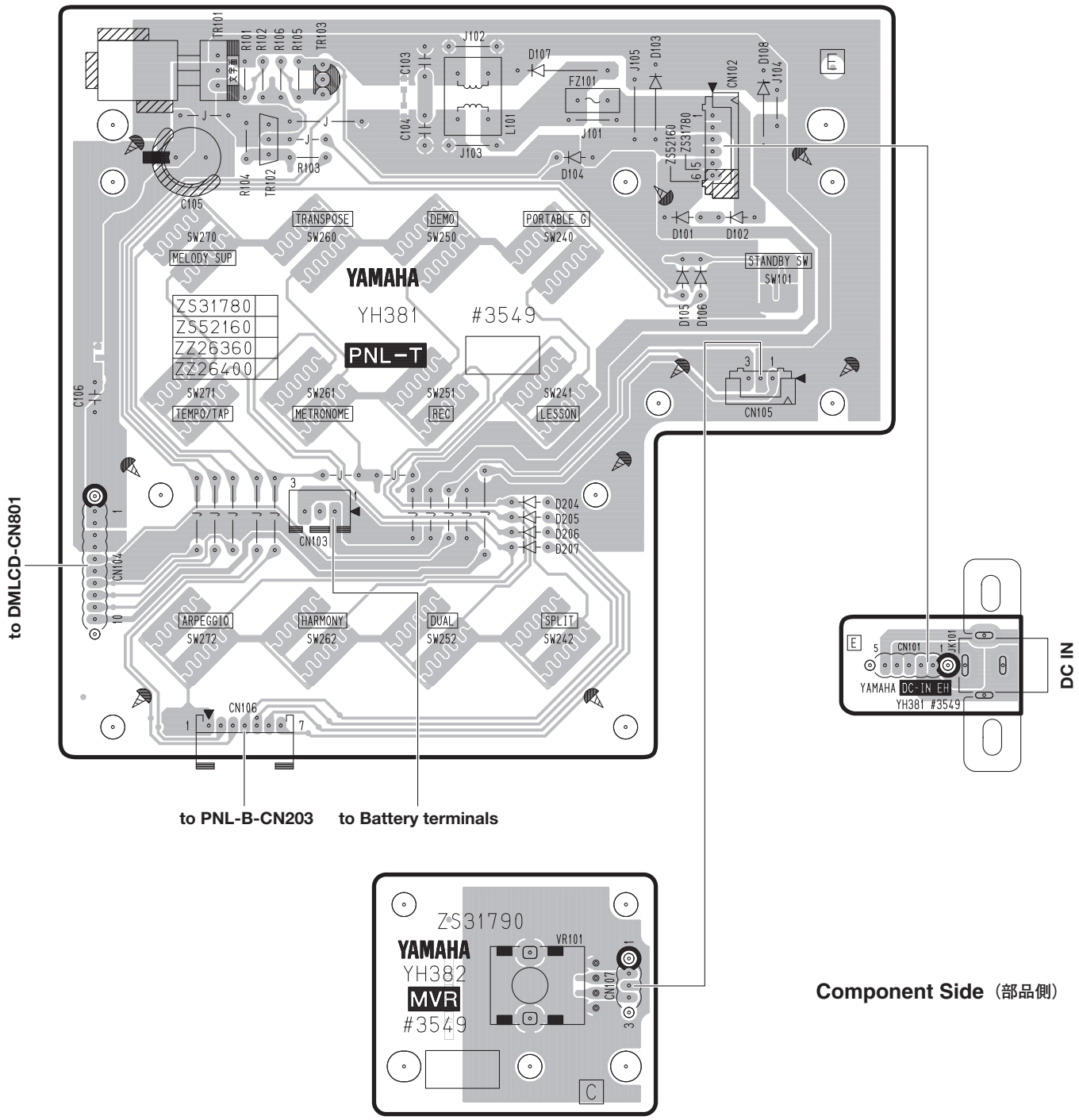
## • ENC Circuit Board




**Component Side**（部品側）

DMLCD: 2NA-ZY33790
ENC: 2NA-ZZ26360

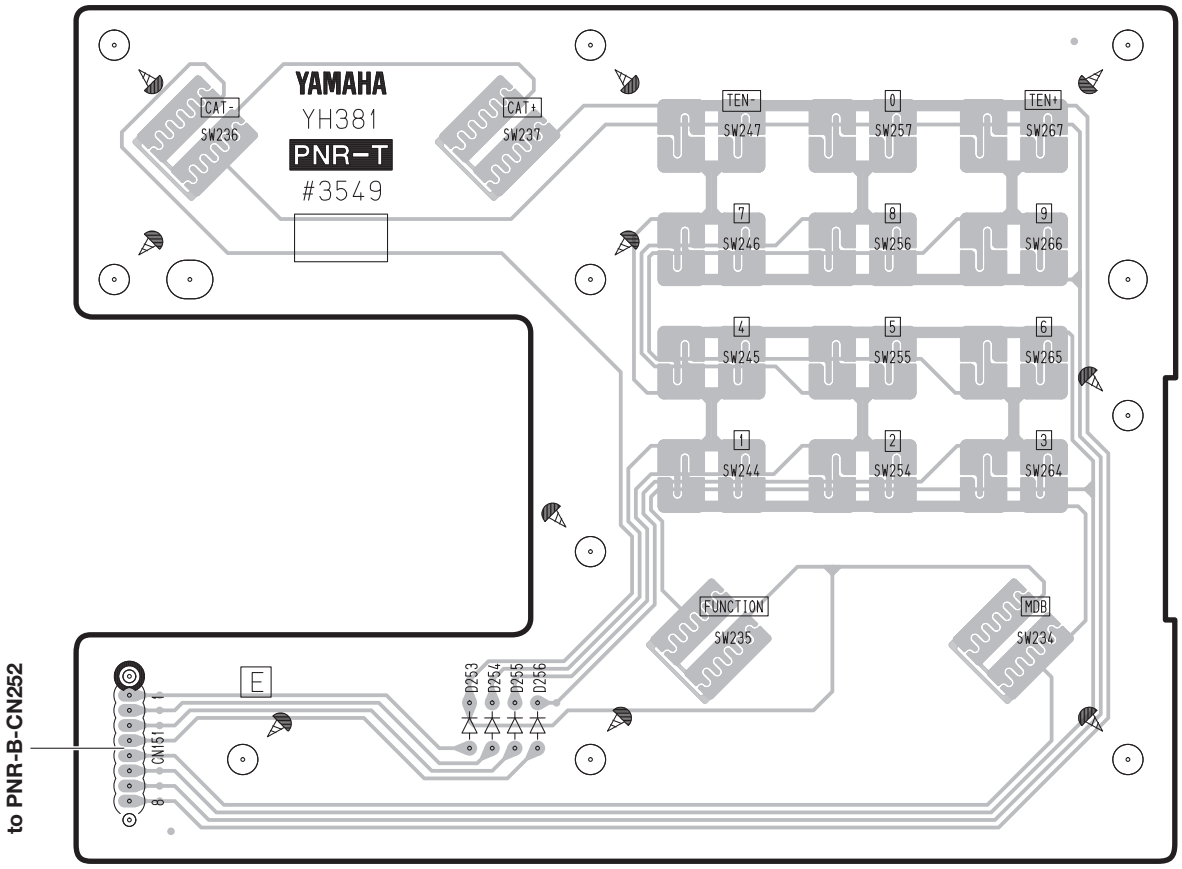
## • PNL-T/MVR/DC-IN EH Circuit Board




**Component Side**（部品側）

DC-IN EH, PNL-T: 2NA-ZZ26360
MVR: 2NA-ZS31790 △3

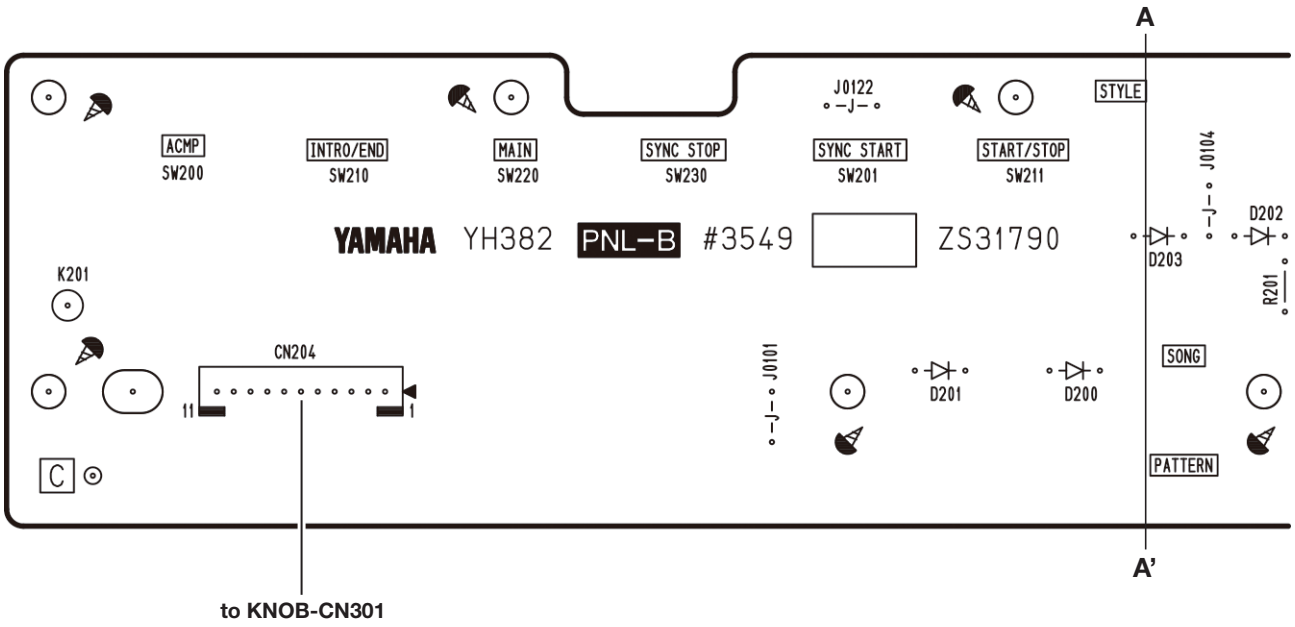
## • PNR-T Circuit Board

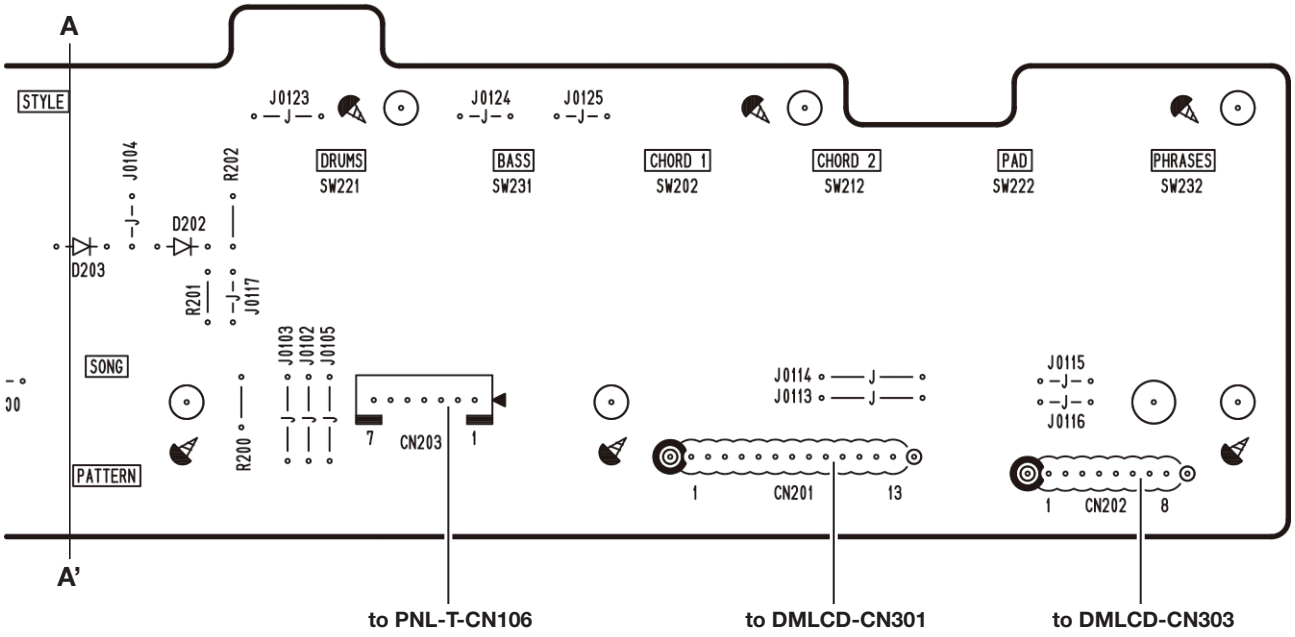



**Component Side**（部品側）

PNR-T: 2NA-ZZ26360

## • PNL-B Circuit Board

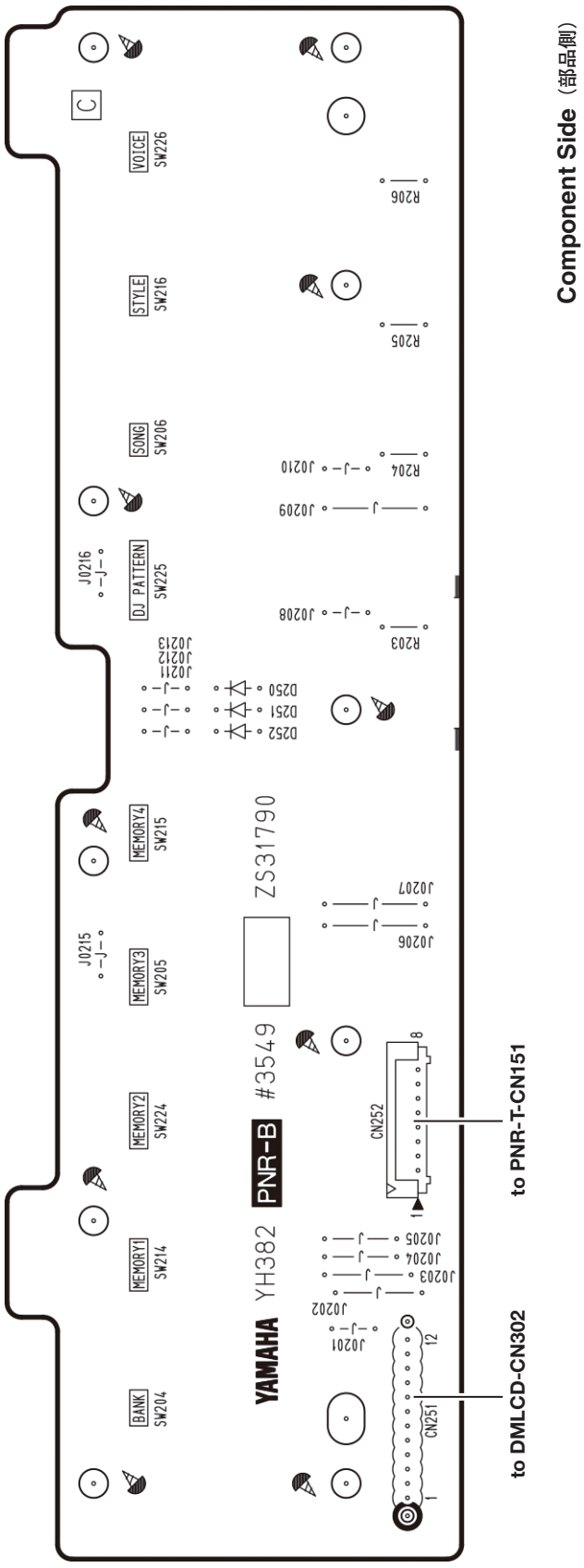



**Pattern Side**（パターン側）

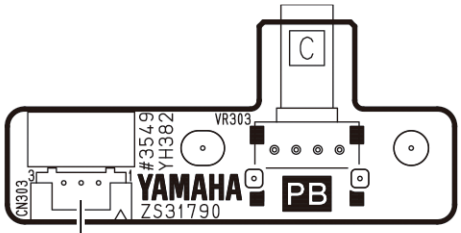
PNL-B: 2NA-ZS31790 3

## • PNR-B Circuit Board

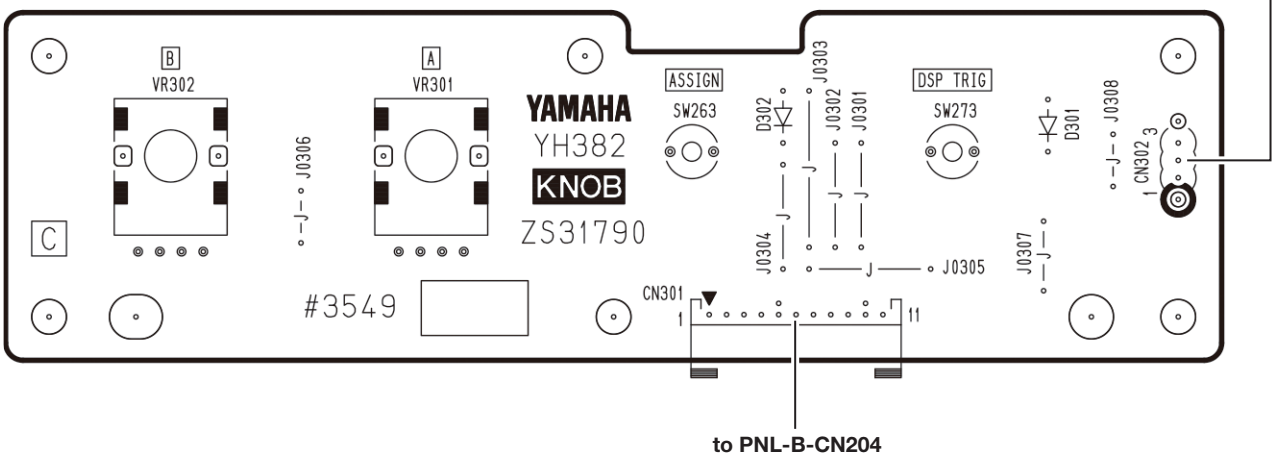



PNR-B: 2NA-ZS31790 ⚠3

## • PB Circuit Board




## • KNOB Circuit Board

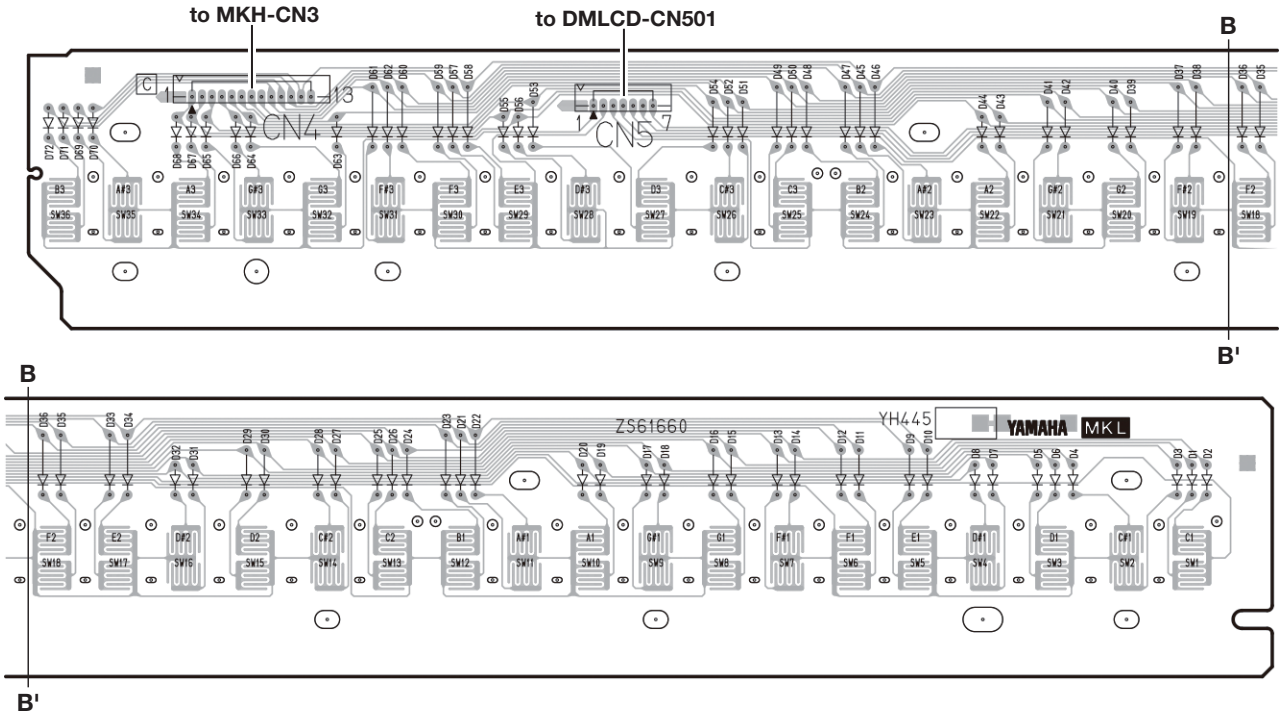



to PNL-B-CN204

**Component Side**（部品側）

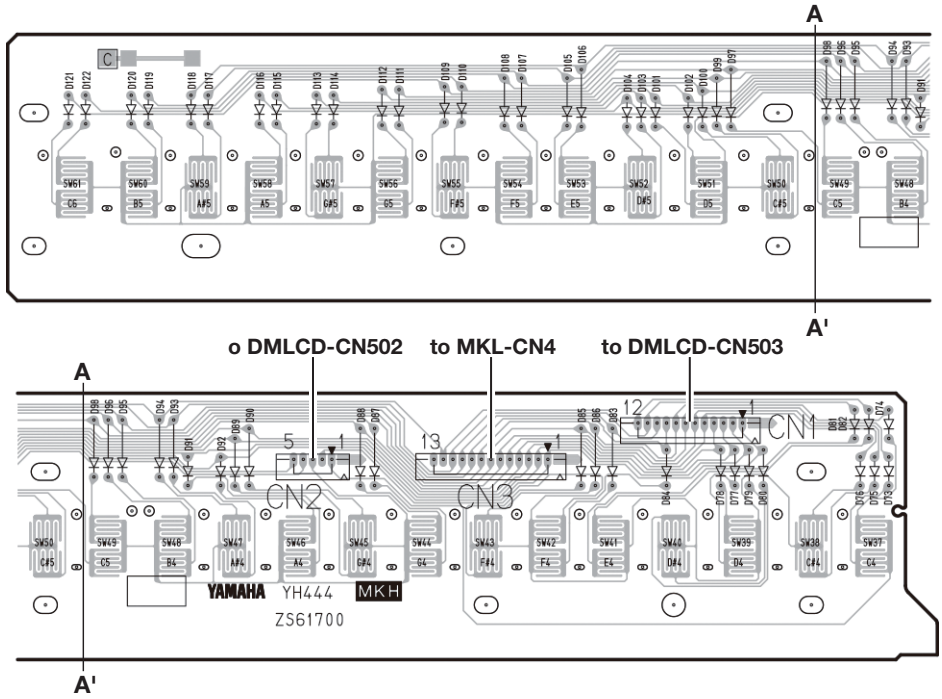
PB, KNOB: 2NA-ZS31790 △3

## • MKL Circuit Board




**Component Side**（部品側）

## • MKH Circuit Board




**Component Side**（部品側）

MKL: 2NAK8-ZS61660
MKH: 2NAK8-ZS61700

# ■ TEST PROGRAM

**CAUTION:**

- **The Test No. 064 “Factory Set” will delete all setup data (included the data that the customer has purchased). Be sure to save the important data to a computer via Musicsoft Downloader software, or tell this to the customer in advance.**

## 1. Measurement condition

### 1-1. Environment

Carry out the tests under the following conditions:

- Ordinary temperature (5°C to 40°C)
- Ordinary humidity (30% to 90%)

### 1-2. Power source

Use an AC adapter: PA-150U or PA-150B.

The power source should be ±10% of the rated voltage of the AC adapter, and should also be its rated frequency.

### 1-3. Measuring instruments

- Frequency counter which can detect thousandth value or more (for check of the output pitch)
- Level meter (using low pass filter fc=15 kHz) (for check of the output level)
- Oscilloscope (for check of the popping noise level)

* Input impedance of the measuring device should be 1 MΩ or more.

### 1-4. Jigs

- Foot switch (FC-4A, FC-5 or an equivalent) (for check of the [SUSTAIN] jack)
- USB cable (A-B type) (for check of the [USB TO HOST] terminal)
- USB flash drive (for check of the [USB TO DEVICE] terminal)
- Oscillator (for inputting a sine wave to the [AUX IN] jack)
- Stereo mini plug cable (for check of the [AUX IN] jack)

### 1-5. Jack connections

[PHONES/OUTPUT] jack: Install a measuring instrument (L, R ch: 33Ω load)

[SUSTAIN] jack: Connect a Foot switch before turning the power ON

### 1-6. Control condition

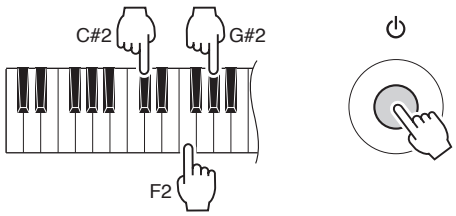
MASTER VOLUME: MAX

Others: the initial setup at turning the power ON

## 2. Test Mode

### 2-1. Starting Test mode

1) While holding down the keys [C#2], [F2] and [G#2] simultaneously, press the [⏻] (Standby/On) switch.




2) When the test program is started, the “TEST” message will appear on the LCD.

### 2-2. Selection/execution of Test items

1) Press the [+] or [-] button in Number buttons area.
2) Press the [START/STOP] button. Refer to the “Test Program Item List” regarding how to operate each test item.
3) To confirm whether the selected test is completed or not:
   Check whether a cursor (“_”) is shown or not below the first character of the test item. If “_” is shown, the selected test has been executed once.

Exit from the current test.

Execute the following operations depending on “OK” or “NG” message.

### 2-3. If the test result shows “OK”

Press the [START/STOP] button, the [DEMO] button, or the lowest key (White Key C1) to return to the item selection display.

### 2-4. If the test result shows “NG”

Press the [DEMO] button or the lowest key (white key C1) to return to the item selection display.

* Note that the [START/STOP] button is ineffective in case of NG.

### 2-5. Test Program Item List

| Number | Test Program Item | Instructions (corresponding to "Selection/execution of Test items") |
|---|---|---|
| 001 | ROM version | Indicates the version written to the ROM.<br>1) Press the [START/STOP] button to call up the Main Program Version as follows:<br>"**Main *****" (***: Version)<br>2) Press the [START/STOP] button to return to the test program selection display. |
| 002 | Memory Check1 All | Executes the simplified check of all the memories (003-005) at one time.<br>1) Press the [START/STOP] button.<br>2) Check the test result.<br>If no problem is found, **"Mem1 OK"** is shown on the LCD.<br>If any problem is found, this test stops at that timing, then **"NG"** is shown on the LCD with the memory name which has a problem as follows:<br>**"Rom NG"**<br>**"Ram NG"**<br>**"From NG"**<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 003 | ROM Check1 | Executes the simplified check of the ROM connected to the CPU bus.<br>1) Press the [START/STOP] button.<br>2) Check the test result.<br>If no problem is found, "**Rom OK**" is shown on the LCD.<br>If any problem is found, "**Rom NG**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 004 | RAM Check1 | Executes the simplified check of the RAM connected to the CPU bus.<br>1) Press the [START/STOP] button.<br>2) Check the test result.<br>If no problem is found, "**Ram OK**" is shown on the LCD.<br>If any problem is found, "**Ram NG**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 005 | Flash ROM Check1 | Executes the simplified check of the Flash ROM connected to the CPU bus.<br>1) Press the [START/STOP] button.<br>2) Check the test result.<br>If no problem is found,"**FRom OK**" is shown on the LCD.<br>If any problem is found, "**FRom NG**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 009 | Pitch Check | Checks whether the correct pitch is output from this instrument or not.<br>1) Before pressing the [START/STOP] button, connect the Frequency Counter to the [PHONES/OUTPUT] jack on the rear panel. (Either L or R ch)<br>2) Press the [START/STOP] button to produce the A3 sound.<br>3) Check the Frequency counter indication.<br>No problem: the indication is with in 441.00±0.2Hz.<br>4) Press the [START/STOP] button to stop the A3 sound.<br>The LCD screen returns to the test program selection display. |
| 011 | Output level R Check | Checks whether the stereo R channel sound is output from the [PHONES/OUTPUT] jack properly or not.<br>1) Before starting the test, connect the level meter to the [PHONES/OUTPUT] jack on the rear panel. (Both L and R ch, 33Ω load)<br>2) Set the MASTER VOLUME to the maximum position.<br>3) Press the [START/STOP] button to produce the 1KHz sine wave sound.<br>4) Check the output levels of L and R ch indicated on the Level meter.<br>No problem if the following conditions are satisfied.<br>[PHONES/OUTPUT] (33Ω load)<br>• R: –6.5dBu±2dB<br>• L: –64.0dBu or less<br>5) Press the [START/STOP] button to stop the 1KHz sine wave sound.<br>The LCD screen returns to the test program selection display. |

| Number | Test Program Item | Instructions (corresponding to "Selection/execution of Test items") |
|---|---|---|
| 012 | **Output level L Check** | Checks whether the stereo L channel sound is output from the [PHONES/OUTPUT] jack properly or not.<br>1) Before pressing the [START/STOP] button, connect the level meter to the [PHONES/OUTPUT] jack on the rear panel. (Both L and R ch, 33Ω load)<br>2) Set the MASTER VOLUME to the maximum position.<br>3) Press the [START/STOP] button to produce the 1KHz sine wave sound.<br>4) Check the output levels of L and R ch indicated on the Level meter.<br>No problem if the following conditions are satisfied.<br>[PHONES/OUTPUT] (33Ω load)<br>• L: -6.5 dBu±2 dB<br>• R: -64.0 dBu or less<br>5) Press the [START/STOP] button to stop the 1KHz sine wave sound.<br>The LCD screen returns to the test program selection display. |
| 018 | **SP MUTE Check** | Checks whether the Speaker MUTE function works properly or not.<br>1) Press the [START/STOP] button.<br>2) Ensure that "**SP OK**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display.<br>* For this test, do not connect anything to the [PHONES/OUTPUT] jack. |
| 019 | **MUTE Check** | Checks whether the Mute function works properly or not.<br>1) Press the [START/STOP] button.<br>The C5 sound is produced, and "**MUTE Off**" is shown on the LCD.<br>2) Press the [+] button to activate the Muting Circuit.<br>The LCD switches to "**MUTE On**".<br>3) Ensure that the speakers as well as the output of the [PHONES/OUTPUT] jack are muted.<br>4) Press the [-] button to cancel the Muting Circuit.<br>The LCD switches to "**MUTE Off**".<br>5) Ensure that the speakers as well as the output of the [PHONES/OUTPUT] jack are not muted.<br>6) Press the [START/STOP] button to stop the C5 sound and return to the test program selection display.<br>* When checking the output of the speakers, ensure that nothing is plugged into the [PHONES/OUTPUT] jack. |
| 021 | **HP Insertion and Extraction Check** | Checks whether headphone insertion and extraction detection is operating properly or not.<br>1) Press the [START/STOP] button.<br>"**HP In**" is shown on the LCD.<br>2) Insert a headphone plug into the [PHONES/OUTPUT] jack.<br>If the insertion is detected, "**HP Out**" is shown on the LCD.<br>3) Remove the headphone plug from the [PHONES/OUTPUT] jack.<br>4) Ensure that the removal is detected and that "**HP OK**" is shown on the LCD.<br>5) Press the [START/STOP] button to return to the test program selection display.<br>* If insertion and extraction of a headphone plug cannot be detected within 60 seconds of the start of the test, the test times out and "**HP NG**" is shown. |

| Number | Test Program Item | Instructions (corresponding to "Selection/execution of Test items") |
|---|---|---|
| 022 | **AUX Check** | Checks whether the sound is input to the [AUX IN] jack properly or not.<br>1) Press the [START/STOP] button.<br>2) Check whether **"AUX In"** is shown on the LCD or not when no plug is plugged into the [AUX-IN] jack.<br>3) Connect an oscillator to the [AUX IN] jack.<br>**"AUX Out"** is shown on the LCD when an stereo mini plug cable is plugged into the [AUX IN] jack properly.<br>4) Connect the level meter to the [PHONES/OUTPUT] jack on the rear panel. (Both L and R ch, 33Ω load)<br>5) Input the sine wave (-4.0dBu, 1 kHz) to the R ch of the [AUX IN] jack.<br>6) Check the output levels of L and R ch indicated on the Level meter.<br>No problem if the following conditions are satisfied.<br>[PHONES/OUTPUT] (33Ω load)<br>• R (input side): -3.0dBu dB<br>• L (non-input side): -60.0 dBu or less<br>7) For the L ch, make the same operations as in Steps 5) to 6).<br>8) Unplug the stereo mini cable from the [AUX IN] jack.<br>If no problem is found, "**AUX End**" is shown on the LCD.<br>9) Press the [START/STOP] button to return to the test program selection display. |
| 024 | **SW/LED Check** | Checks whether each panel button works properly or not.<br>1) Press the [START/STOP] button.<br>2) "**Dial Up**" is shown on the LCD.<br>3) Turn the dial in the Up direction (to the right).<br>If the Up-direction dial movement is detected, "**Dial Dwn**" is shown on the LCD.<br>4) Turn the dial in the Down direction (to the left).<br>If the Down-direction dial movement is detected, "**Sampling**" is shown on the LCD.<br>5) Press the buttons that were shown earlier in order.<br>The sound of the note assigned to the pressed button will be produced.<br>Regarding what note is assigned, refer to the "switch test item list."<br>6) Press the [START/STOP] button to return to the test program selection display.<br>* You can exit from this test by pressing the [DEMO] button or the lowest key (white key C1).<br>* If any problem is found, "**SW NG**" is shown on the LCD.<br>* If two or more buttons are pressed simultaneously, "**Over Two**" is shown on the LCD. |
| 025 | **All LED On** | Checks whether all the LED lamps turned on properly or not.<br>1) Press the [START/STOP] button.<br>2) Check whether all of the LED lamps light up.<br>3) Press the [START/STOP] button to turn off the LED lamps and return to the test program selection display. |
| 031 | **All LCD On** | Checks whether all the LCD dots are turned on properly or not.<br>1) Press the [START/STOP] button.<br>2) Check that all the LCD dots are turned on.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 032 | **All LCD Off** | Checks whether all the LCD dots are turned off properly or not.<br>1) Press the [START/STOP] button.<br>2) Check that all the LCD dots are turned off.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 034 | **Main Volume Check** | Checks whether the [MASTER VOLUME] control works properly or not.<br>1) Press the [START/STOP] button.<br>"**MVol Min**" is shown on the LCD.<br>2) Set the [MASTER VOLUME] control to the minimum.<br>If the minimum value is detected, "**MVol Max**" is shown on the LCD.<br>3) Set the [MASTER VOLUME] control to the maximum.<br>4) Ensure that the maximum value is detected and that "**MVol OK**" is shown on the LCD.<br>5) Press the [START/STOP] button to return to the test program selection display. |

| Number | Test Program Item | Instructions (corresponding to "Selection/execution of Test items") |
|---|---|---|
| 035 | Fail Safe Check | Checks whether the Fail-Safe Circuit of the Digital Volume works properly or not.<br>1) Press the [START/STOP] button.<br>2) Ensure that "**FS OK**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 036 | Pedal1 Check | Checks whether the Foot switch plugged into the [SUSTAIN] jack works properly or not.<br>1) Press the [START/STOP] button to produce the C3 sound.<br>2) Press the Foot switch to produce the C4 sound.<br>3) Release the Foot switch to stop the C4 sound.<br>If no problem is found, "**PD1 OK**" is shown on the LCD.<br>4) Press the [START/STOP] button to return to the test program selection display.<br>* Connect a Foot switch to the [SUSTAIN] jack before turning the power ON. |
| 040 | Pitch Bend Check | Checks whether the [PITCH BEND] wheel control works properly or not.<br>1) Press the [START/STOP] button.<br>"**PB UP**" is shown on the LCD.<br>2) Set the [PITCH BEND] wheel to the maximum.<br>Ensure that the maximum value is detected, and that "**PB DW**" is shown on the LCD and the G3 sound is produced.<br>3) Set the [PITCH BEND] wheel to the minimum.<br>Ensure that the minimum value is detected, and that "**PB C**" is shown on the LCD and the C3 sound is produced.<br>4) Return the [PITCH BEND] wheel to the center.<br>Ensure that the center value is detected, and that "**PB OK**" is shown on the LCD and the C4 sound is produced.<br>5) Press the [START/STOP] button to stop the C4 sound and return to the test program selection display. |
| 041 | Knob Check | Checks whether the [LIVE CONTROL A, B] knobs control works properly or not.<br>1) Press the [START/STOP] button.<br>"**Knob1 Lo**" is shown on the LCD.<br>2) Set the [LIVE CONTROL A] knob to the minimum.<br>If the minimum value is detected, the C3 sound is produced, and "**Knob1 Hi**" is shown on the LCD.<br>3) Set the [LIVE CONTROL A] knob to the maximum.<br>If the maximum value is detected, the C4 sound is produced, and "**Knob1 C**" is shown on the LCD.<br>4) Set the [LIVE CONTROL A] knob to the center.<br>If the center value is detected, the C4 sound stops, and "**Knob2 Lo**" is shown on the LCD.<br>5) Set the [LIVE CONTROL B] knob to the minimum.<br>If the minimum value is detected, the C3 sound is produced, and "**Knob2 Hi**" is shown on the LCD.<br>6) Set the [LIVE CONTROL B] knob to the maximum.<br>If the maximum value is detected, the C4 sound is produced, and "**Knob2 C**" is shown on the LCD.<br>7) Set the [LIVE CONTROL B] knob to the center.<br>It the center value is detected, the C4 sound stops and "**Knob OK**" is shown on the LCD.<br>8) Press the [START/STOP] button to return to the test program selection display. |

| Number | Test Program Item | Instructions (corresponding to "Selection/execution of Test items") |
|---|---|---|
| 044 | **USB Connection Check** | Checks whether both the [USB TO DEVICE] terminal and the [USB TO HOST] terminal are working properly or not.<br>1) First, press the [START/STOP] button.<br>"**Conn --**" is shown on the LCD.<br>2) Connect a USB cable between the [USB TO DEVICE] terminal and the [USB TO HOST] terminal.<br>3) Ensure that "**Conn OK**" is shown on the LCD and that the C4 sound is produced.<br>* If a connection cannot be detected within 60 seconds of the start of the test, the test times out and "**Conn NG**" is shown.<br>4) Press the [START/STOP] button to stop the C4 sound and return to the test program selection display.<br>5) Disconnect the USB cable. |
| 045 | **USB Storage Check** | Checks whether the USB storage equipment is operating properly or not.<br>1) Insert a USB flash drive into the [USB TO DEVICE] terminal.<br>2) Press the [START/STOP] button.<br>3) Check that "**Strg OK**" is shown on the LCD.<br>*"**Strg --**" is shown while proper operation is being confirmed.<br>*If no media is inserted, "**Strg no**" is shown.<br>*If protected media is inserted, "**Strg Prt**" is shown.<br>*If a read/write failure occurs, "**Strg NG**" is shown.<br>4) Press the [START/STOP] button to return to the test program selection display.<br>5) Remove the USB flash drive. |
| 051 | **Ping Pong Mode Check** | Checks whether the two contacts of each key are operating properly or not.<br>1) Press the [START/STOP] button.<br>"**Press**" is shown on the LCD.<br>2) Slowly press the key that you want to check.<br>3) When the first contact turns ON, "**Make 1/note number**" is shown, and the sound of the pressed key is produced.<br>4) When the second contact turns ON, "**Make 1 2/note number**" is shown, and the sound of the pressed key +4 is produced.<br>5) When the key is released and the second contact turns OFF, the sound is stopped, and when the first contact turns OFF, "**Press**" is shown on the LCD.<br>6) Repeat steps 2) to 5) for the rest of the keys to be checked.<br>7) Press the [START/STOP] button to return to the test program selection display.<br>* In this test item, pressing the lowest key will not return to the test program selection display. |
| 058 | **ROM Check2** | Executes a check of the ROM connected to the CPU bus. (Full address)<br>1) Press the [START/STOP] button.<br>2) Check the result. (This check takes about 1 minutes to complete.)<br>If no problem is found, "**Rom OK**" is shown on the LCD.<br>If any problem is found, "**Rom NG**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 059 | **RAM Check2** | Executes a check of the RAM connected to the CPU bus (Full address).<br>1) Press the [START/STOP] button.<br>2) Check the test result.<br>If no problem is found, "**Ram OK**" is shown on the LCD.<br>If any problem is found, "**Ram NG**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 060 | **Flash ROM Check2** | Executes the check of the Flash ROM connected to the CPU bus (Full address).<br>1) Press the [START/STOP] button.<br>2) Check the result. (This check takes about 1 minutes 40 seconds to complete.)<br>If no problem is found, "**FRom OK**" is shown on the LCD.<br>If any problem is found, "**FRom NG**" is shown on the LCD.<br>3) Press the [START/STOP] button to return to the test program selection display. |

| Number | Test Program Item | Instructions (corresponding to "Selection/execution of Test items") |
|---|---|---|
| 063 | **Keyboard Check** | Checks whether each key works properly or not.<br>1) Press the [START/STOP] button.<br>2) Check that the LCD shows you the should-be-pressed key.<br>3) Press the key specified on the LCD one by one.<br>The each sound between C3 and B4 assigned to the pressed key will be produced.<br>4) Press the [START/STOP] button to return to the test program selection display.<br>* You can exit from this test by pressing the [DEMO] button.<br>* If the wrong key is pressed, nothing happens.<br>* If two or more keys are pressed simultaneously, "**Over Two**" is shown on the LCD.<br>* If the velocity of pressing the key is not suitable, either "**Too Slow**" or "**Too Fast**" is shown on the LCD. |
| 064 | **Factory Set** | Reset all the backup region of the memories to the initial factory status.<br>1) Press the [START/STOP] button.<br>2) Do not turn off the power while "**Fact --**" is shown on the LCD.<br>After initialization is completed, "**Fact End**" is shown.<br>3) Press the [START/STOP] button to return to the test program selection display. |
| 065 | **Test Exit** | Lets you exit from the Test mode to the normal mode.<br>1) Press the [START/STOP] button.<br>The Test Program mode will end, and the PSR-E463 will be restarted.<br>Never turn off the power until the Main display appears. Doing so may cause a malfunction. |

### 3. Other Test Items (Check in Normal Mode)

| No. | Test Item | Instruction |
|---|---|---|
| - | **Popping Noise Level Check** | Checks whether the popping noise level is proper or not when turning the power on and off.<br>1) Connect an Oscilloscope to the [PHONES/OUTPUT] jack.<br>2) Set the MASTER VOLUME to the maximum position.<br>3) Press the [STANDBY/ON] switch to check the noise level indicated on the Oscilloscope when turning the power of the PSR-E463 on and off.<br>If the value is within the range below, the specifications are satisfied.<br>[PHONES/OUTPUT] (Both L and R ch, 33Ω load)<br>• L, R: 1.0V peak to peak or less |
| - | **Noise Level Check** | Checks whether the noise level is proper or not.<br>1) Connect the level meter to the [PHONES/OUTPUT] jack on the rear panel. (Both L and R ch, 33Ω load)<br>2) Set the MASTER VOLUME to the maximum position.<br>3) Check the output levels of L and R ch indicated on the Level meter.<br>[PHONES/OUTPUT] (33Ω load)<br>• L, R: –75 dBu or less<br>* Do not plug any plugs into the [AUX IN] jack. |
| - | **No charging the batteries Check** | Checks whether voltage is supplied to the batteries or not.<br>1) Connect the AC adapter to the [DC IN] jack.<br>2) Check the voltage between battery terminal A and B.<br>No problem if the voltage is 0.2V or lower. |
| - | **Vibration Noise Check** | Checks whether vibration noise occurs on the PSR-E463 or not.<br>1) Select the Voice following: No. 028 (16+2 Org).<br>2) Set the Main Voice volume in the Function to the default value.<br>3) Set the MASTER VOLUME to the maximum position.<br>4) Carry out the scaling to check whether vibration noise occurs or not.<br>No problem if no vibration noises occur. |

## 4. Test Program List

| Turn | SW Name | LCD Display | Note Number |
|---|---|---|---|
| 1 | DIAL UP | Dial Up | C2 |
| 2 | DIAL DOWN | Dial Dwn | C#2 |
| 3 | QUICK SAMPLING | Sampling | D2 |
| 4 | DEMO | Demo | D#2 |
| 5 | TRANSPOSE | Transpos | E2 |
| 6 | MELODY SUPPRESSOR | MELODY | F2 |
| 7 | AUDIO | Audio | F#2 |
| 8 | REC | REC | G2 |
| 9 | METRONOME | Metro | G#2 |
| 10 | TEMPO/TAP | TAP | A2 |
| 11 | SPLIT | Split | A#2 |
| 12 | DUAL | Dual | B2 |
| 13 | HARMONY/ARPEGGIO | Harmony | C3 |
| 14 | DSP | DSP | C#3 |
| 15 | CATEGORY << | Catego << | D3 |
| 16 | CATEGORY >> | Catego >> | D#3 |
| 17 | FUNCTION | Function | E3 |
| 18 | PORTABLE GRAND | Piano | F3 |
| 19 | Tenkey 1 | Tenkey 1 | F#3 |
| 20 | Tenkey 2 | Tenkey 2 | G3 |
| 21 | Tenkey 3 | Tenkey 3 | G#3 |
| 22 | Tenkey 4 | Tenkey 4 | A3 |
| 23 | Tenkey 5 | Tenkey 5 | A#3 |
| 24 | Tenkey 6 | Tenkey 6 | B3 |
| 25 | Tenkey 7 | Tenkey 7 | C4 |
| 26 | Tenkey 8 | Tenkey 8 | C#4 |
| 27 | Tenkey 9 | Tenkey 9 | D4 |
| 28 | Tenkey - | Tenkey - | D#4 |
| 29 | Tenkey 0 | Tenkey 0 | E4 |
| 30 | Tenkey + | Tenkey + | F4 |
| 31 | TARGET CHANGE | Target | F#4 |
| 32 | ASSIGN | Assign | G4 |
| 33 | ACMP | ACMP | G#4 |
| 34 | INTRO/ENDING/rit. | Intro | A4 |
| 35 | MAIN / AUTO FILL | Main/Fil | A#4 |
| 36 | SYNC STOP | S.Stop | B4 |
| 37 | SYNC START | S.Start | C5 |
| 38 | START / STOP | Str/Stp | C#5 |
| 39 | SONG / AUDIO 1 | Song 1 | D5 |
| 40 | SONG / AUDIO 2 | Song 2 | D#5 |
| 41 | SONG / AUDIO 3 | Song 3 | E5 |
| 42 | SONG / AUDIO 4 | Song 4 | F5 |
| 43 | SONG / AUDIO 5 | Song 5 | F#5 |
| 44 | SONG / AUDIO A | Song A | G5 |
| 45 | BANK | Memory | G#5 |
| 46 | REGIST 1 | Regist 1 | A5 |
| 47 | REGIST 2 | Regist 2 | A#5 |
| 48 | REGIST 3 | Regist 3 | B5 |
| 49 | REGIST 4 | Regist 4 | C6 |
| 50 | VOICE | Voice | C#6 |
| 51 | STYLE | Style | D6 |
| 52 | SONG | Song | D#6 |
| 53 | GROOVE CREATOR | Grooue | E6 |

# ■ DMLCD CIRCUIT BOARD CHECK METHOD

The DMLCD Circuit Board is provided with test points for service check purposes.
Check the test points on the DM LCD Circuit Board if the following symptoms appear.

Symptoms and check items
1) No LCD display with Power SW ON --> Check items 1 to 4 sequentially
2) No sound or distorted sound --> Check items 5 to 8 sequentially
Test Point

| NO. | Test Point | Circuit | Judgment criteria | Measured by | Parts with possible defects |
|---|---|---|---|---|---|
| ① | +B | Power supply for DMLCD circuit board | More than 10.0V | Multimeter | TR101 or FZ101 (On PNL-T Circuit Board) |
| ② | +5D | 5V power for digital circuit | 5.0V±0.5V | Multimeter | IC801 |
| ③ | +3.3D | 3.3V power for digital circuit | 3.3V±0.3V | Multimeter | IC802 |
| ④ | RESET | CPU & memory reset signal | 3.3V±0.5V | Multimeter | IC111 |
| ⑤ | +HP | Power for HP circuit | More than 9.0V | Multimeter | TR202 |
| ⑥ | +3.3A | 3.3V power for analog circuit | 3.3V±0.3V | Multimeter | IC802 |
| ⑦ | DAC-L | DAC output L channel | There shall be audio output without distortion. | Signal Checker | IC201 |
| ⑧ | DAC-R | DAC output R channel | There shall be audio output without distortion. | Signal Checker | IC201 |

Note[1]: Use the standard AC adapter PA-150U or PA-150B for check operation.

# ■ DMLCD Circuit Board チェック方法

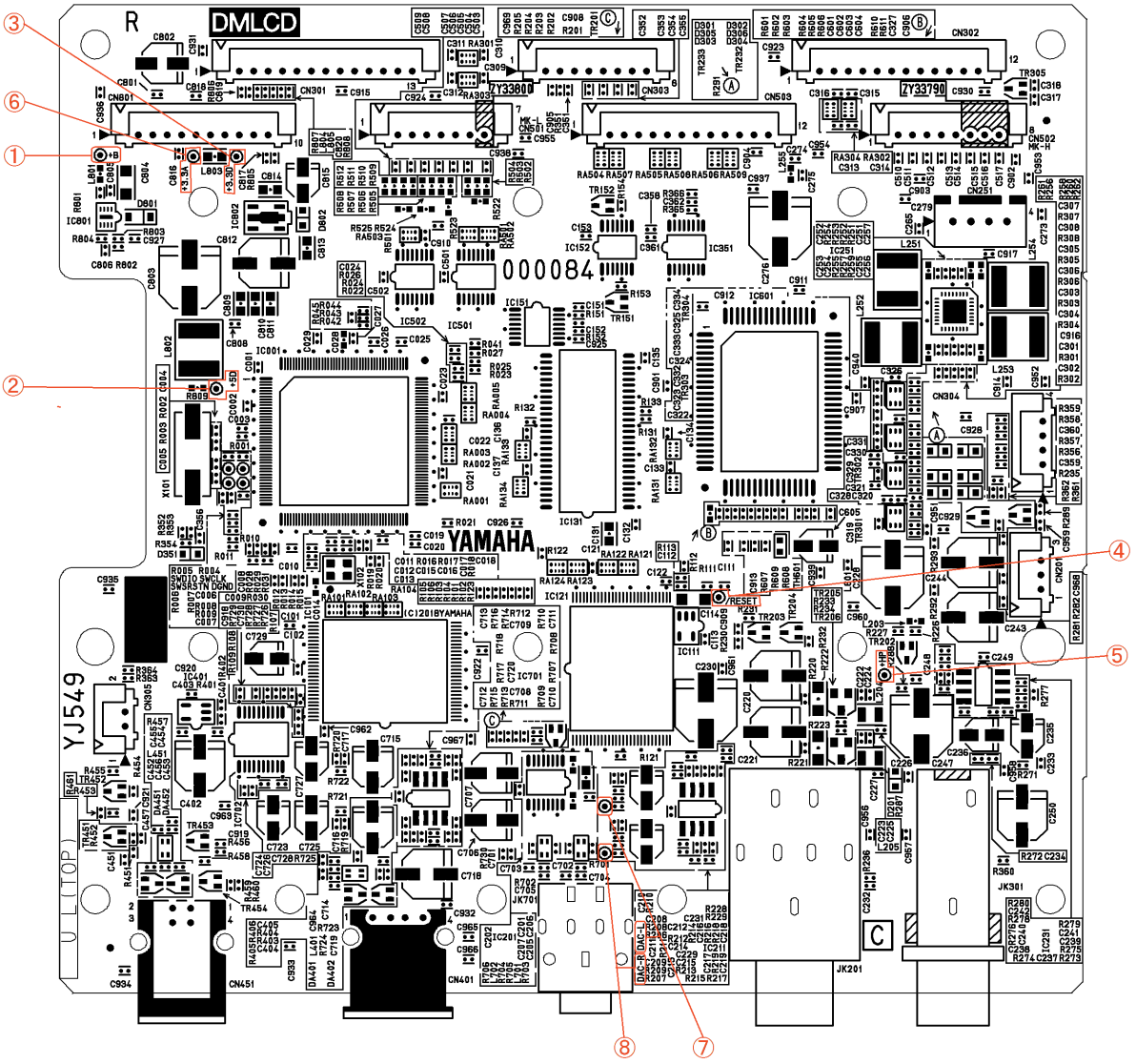
DMLCD Circuit Board にはサービスチェック用の Test Point を設けてあります。
下記の症状により DMLCD Circuit Board の Test Point を確認してください。

症状により以下の箇所を確認する
1) 電源を入れても LCD が点灯しない。⇒ ①～④を順番に確認する。
2) 音が出ない、音が歪む ⇒ ⑤～⑧を順番に確認する。
TEST POINT チェック

| NO. | Test Point | 回路 | 判定基準 | 測定器具 | 不良が疑われる部品 |
|---|---|---|---|---|---|
| ① | +B | DMｼｰﾄ電源 | 10.0V以上 | テスター | TR101 or FZ101 (PNL-Tシート) |
| ② | +5D | デジタル回路用5V電源 | 5.0V±0.5V | テスター | IC801 |
| ③ | +3.3D | デジタル回路用3.3V電源 | 3.3V±0.3V | テスター | IC802 |
| ④ | RESET | CPU,メモリー Reset信号 | 3.3V±0.5V | テスター | IC111 |
| ⑤ | +HP | ヘッドホン回路用5V電源 | 9.0V以上 | テスター | TR202 |
| ⑥ | +3.3A | アナログ回路用3.3V電源 | 3.3V±0.3V | テスター | IC802 |
| ⑦ | DAC-L | DAC出力Lチャンネル | 音声が出力されかつ歪無き事 | Signal Checker | IC201 |
| ⑧ | DAC-R | DAC出力Rチャンネル | 音声が出力されかつ歪無き事 | Signal Checker | IC201 |

注[1]: チェック作業時は標準の AC アダプター PA-150B を使用すること。

## • DMLCD Circuit Board (ZY337900)




Component Side ( 部品側 )

# ■ BACKUP

The following Backup parameters will be maintained even if the power is turned off.

**● Backup parameters (each time)**

- User Songs
- Style data transferred from a computer and loaded to Style numbers 236–245
- Assigned Sample to keyboards

**● Backup parameters when turning off the power**

- Registration Memory
- FUNCTION Settings:
  Style Volume, Song Volume, Groove Volume, Audio Input Volume, Tuning, Split Point, Touch Response, Master EQ Type, Audio Loop Back on/off, Blank Cut on/off, Metronome Volume, Demo Group, Demo Play Mode, Auto Power Off Time, Battery Type, Language

In addition to the Backup parameters above, all the data (including Style data that has not been loaded) transferred from the connected computer will be maintained even if you turn off the power.

### NOTICE

Backup is executed automatically when the power is turned off. When the power is turned off in the following cases, backup is not executed.

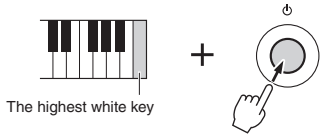
- Disconnected AC adaptor
- Power failure
- Low or no battery power

# ■ INITIALIZATION

You can erase the backup parameter settings described above, and then restore all the default factory settings. The instrument can be initialized with the following two methods.

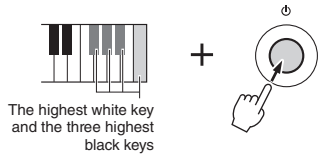
### • Backup Clear

This operation initializes the backup parameters. While holding down the highest white key, press the [⏻] (Standby/On) switch to turn the power on.
During the process, ”Clr BackUp” is shown.




### • Flash Clear

This operation deletes all the Songs and Styles that have been transferred from a computer. Note that data loaded to Style numbers 236–245 will be maintained.
While simultaneously holding the highest white key and the three highest black keys, press the [⏻] (Standby/On) switch to turn the power on. During the process, “Clr Flash” is shown.




### NOTICE

- Keep in mind that this operation also deletes any data you have purchased and loaded. Be sure to save important data to a computer. For details, refer to the online PDF manual (page 11) “Computer-related Operations.”

## ■ バックアップ

以下のデータは自動的にバックアップ ( 保存 ) されるため、電源を切ってもデータは消えません。

**● 随時バックアップされる内容**

- ユーザーソング
- コンピューターから転送し、スタイル番号 236 ～ 245 に読み込みされたスタイルファイル
- 鍵盤に登録されたサンプル

**● 電源を切ったときにバックアップされる内容**

- レジストレーションメモリー
- 機能設定の各設定：
  スタイル音量、ソング音量、グルーブ音量、オーディオ入力音量、チューニング、スプリットポイント、タッチレスポンス、マスターEQタイプ、オーディオループバック オン / オフ、ブランクカット オン / オフ、メトロノーム音量、デモグループ、デモ再生モード、オートパワーオフ設定、電池選択、言語

上記のバックアップデータに加え、コンピューターから転送した全スタイルデータ ( 読み込まれていないものも含む ) と全ソングデータは、電源をオフにしても消えずに楽器本体内に保存されます。

### 注記

バックアップは電源を切るときに自動的に実行されますが、下記の状況で電源が切れたときは、バックアップが実行されませんのでご注意ください。

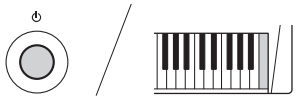
- 電源が入っている状態で、電源アダプターを抜いたとき
- 停電などで突然電源が切れたとき
- 電池を使用していて、消耗して電源が切れたとき

## ■ 初期化

バックアップデータなどを消去し、設定を工場出荷時の状態に戻すことを「 初期化 」と呼びます。目的に応じて、以下の 2 つの方法で初期化を行なってください。

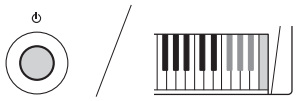
### ● バックアップクリア

バックアップデータを初期状態に戻します。鍵盤の最高音 (白鍵) を押しながら［⏻］(スタンバイ / オン) スイッチを押して電源を入れます。処理中は“Clr BackUp”が表示されます。



### ● フラッシュクリア

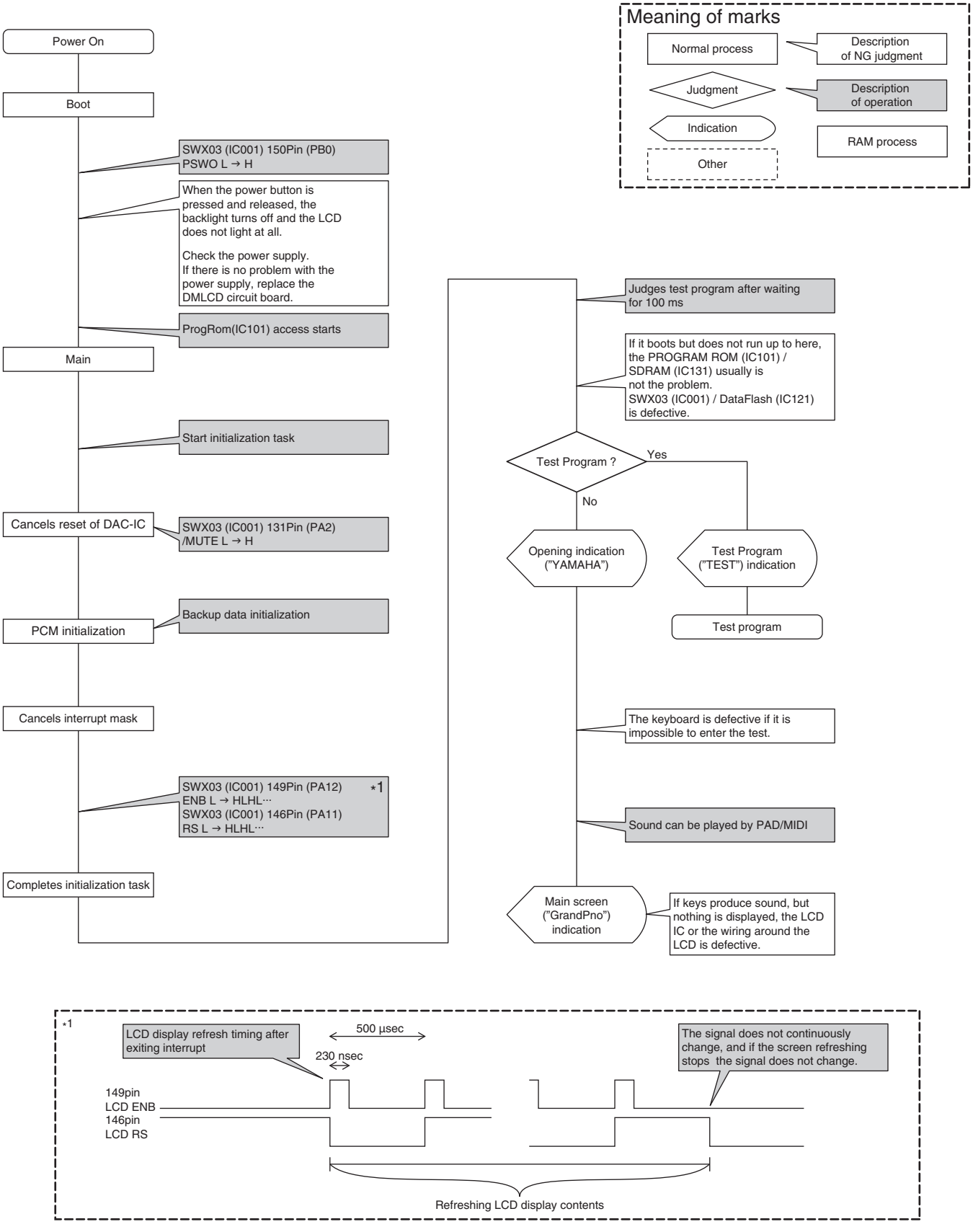
コンピューターから転送したソング / スタイルの全データを消去します。ただし、スタイル番号 236 ～ 245 に読み込まれたデータは消去されません。鍵盤の最高音 ( 白鍵 ) と一番高い黒鍵 3 つを同時に押したまま、［⏻］( スタンバイ / オン ) スイッチを押して電源を入れます。処理中は“Clr Flash”が表示されます。



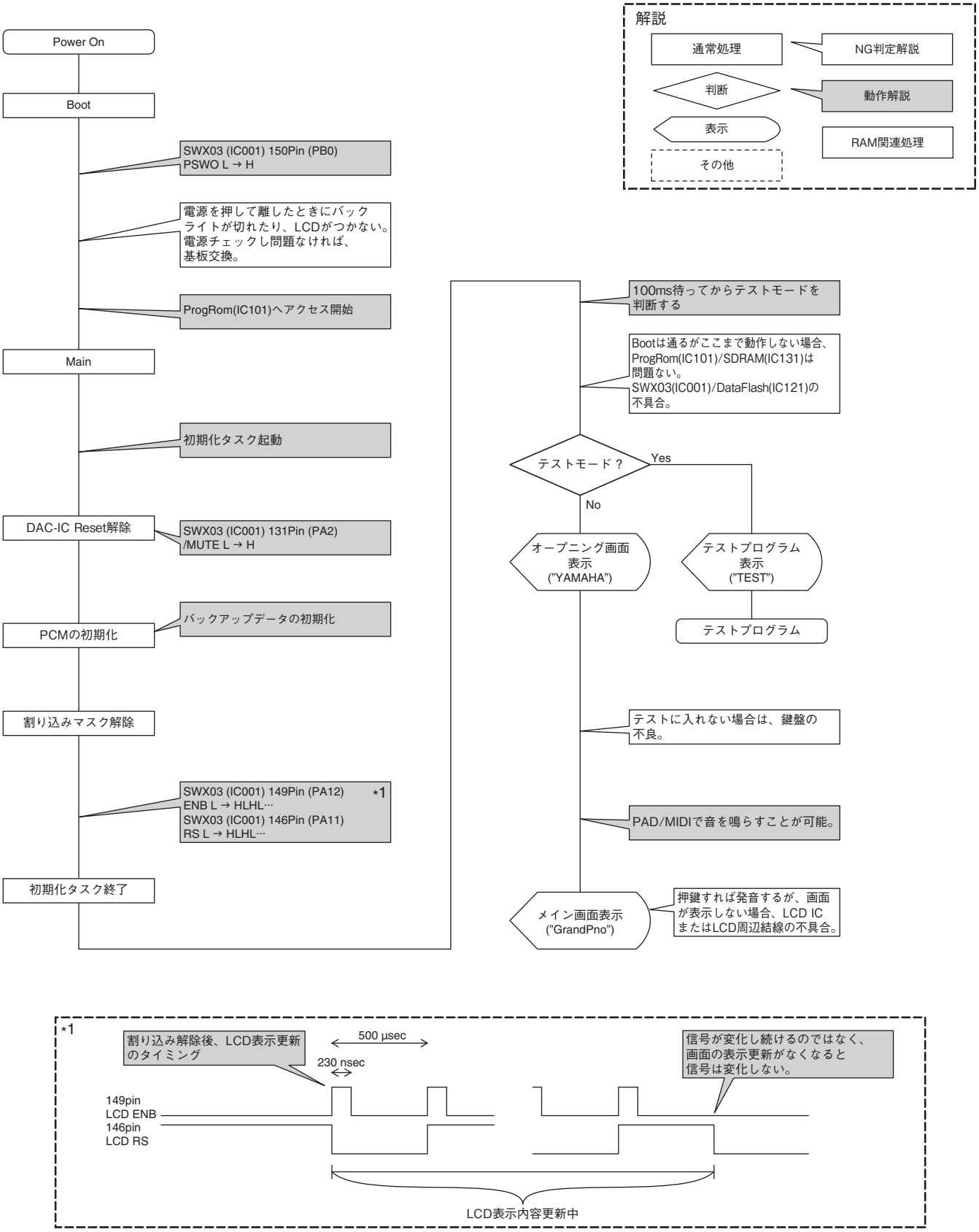
### 注記

- この操作により、購入した有料のデータも消去されます。消去したくないデータは、必要に応じてコンピューターに転送 / 保存します。詳細は、電子マニュアル (11 ページ ) 「コンピューターとつなぐ」をご覧ください。

# ■ SYSTEM BOOTING FLOW CHART



# ■ システム起動フローチャート



PORTATONE

PSR-E463

# PARTS LIST

■ **CONTENTS**（目次）


**OVERALL ASSEMBLY**（総組立）..... **2**
**UPPER CASE ASSEMBLY**（上ケース **Ass'y**）..... **4**
**LOWER KEY BED ASSEMBLY**（下ケース鍵盤 **Ass'y**）..... **6**
**LOWER CASE ASSEMBLY**（下ケース **Ass'y**）..... **7**
**KEYBOARD ASSEMBLY**（**16NS** 鍵盤）..... **8**
**ELECTRICAL PARTS**（電気部品）..... **9-12**


**Notes : DESTINATION ABBREVIATIONS**

| | | | |
|---|---|---|---|
| A: | Australian model | M: | South African model |
| B: | British model | O: | Chinese model |
| C: | Canadian model | P: | Brazilian model |
| D: | German model | Q: | South-east Asia model |
| E: | European model | T: | Taiwan model |
| F: | French model | U: | U.S.A. model |
| H: | North European model | V: | General export model (110V) |
| I: | Indonesian model | W: | General export model (220V) |
| J: | Japanese model | N,X: | General export model |
| K: | Korean model | Y: | Export model |
| | | Z: | Indian model |

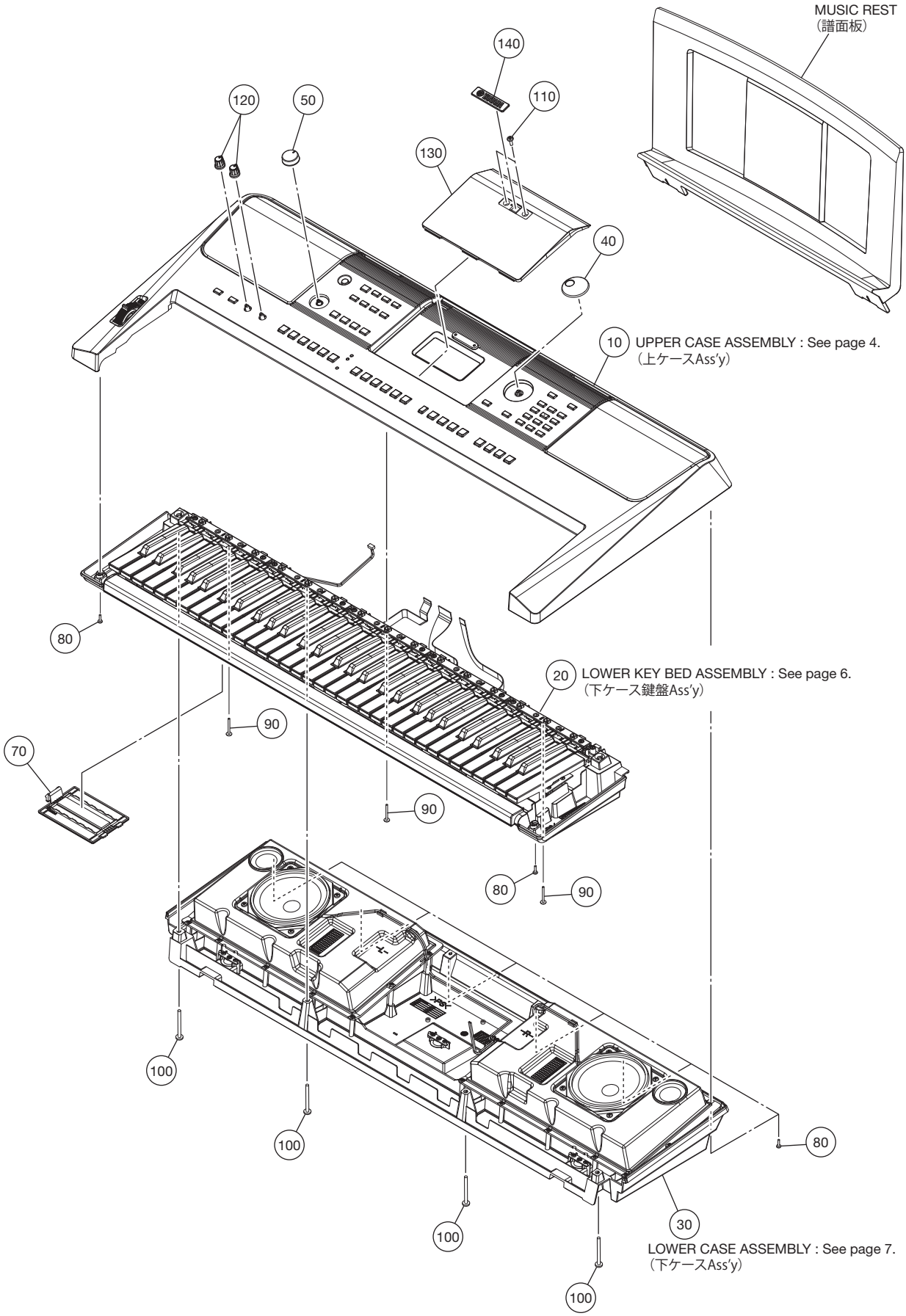
## ■ WARNING

Components having special characteristics are marked ⚠ and must be replaced with parts having specifications equal to those originally installed.

⚠ 印の部品は、安全を維持するために重要な部品です。交換をする場合は、安全のため必ず指定の部品をご使用下さい。

- The numbers “QTY” show quantities for each unit.
- The parts with “--” in “PART NO.” are not available as spare parts.
- This mark “ } ” in the REMARKS column means these parts are interchangeable.
- The second letter of the shaded ( ) part number is O, not zero.
- The second letter of the shaded ( ) part number is I, not one.
- QTY 欄に記されている数字は、各ユニット当たりの使用個数です。
- PART NO. が "--" の部品は、サービス用部品として準備されておりません。
- REMARKS 欄の「 } 」マークの部品は、併用部品です。
- 網掛けの付いた PART NO. の２番目の文字は「ゼロ」ではなく「オー」です。
- 網掛けの付いた PART NO. の２番目の文字は「イチ」ではなく「アイ」です。

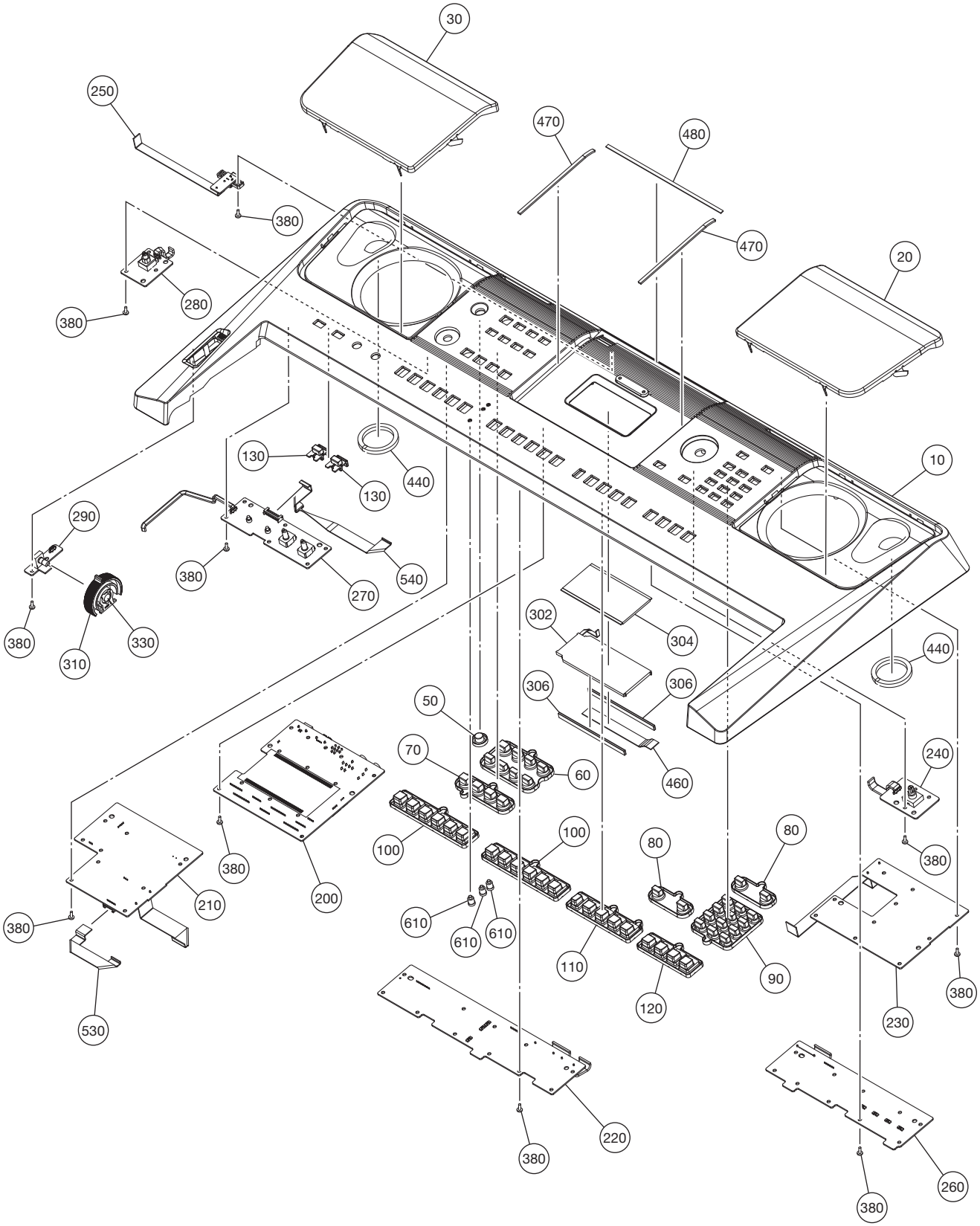
# ■ OVERALL ASSEMBLY（総組立）



| | REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|---|
| | | | OVERALL ASSEMBLY | | 総組立 | PSR-E463 | |
| | | -- | FINAL ASSEMBLY | PK PSR-E463 | 総組立 | (ZZ25740) | |
| | 10 | -- | UPPER CASE ASSEMBLY | PK PSR-E463 | 上ケースAss'y | (ZZ25730) | |
| | 20 | -- | LOWER KEY BED ASSEMBLY | | 下ケース鍵盤Ass'y | (ZS36080) | |
| | 30 | -- | LOWER CASE ASSEMBLY | | 下ケースAss'y | (ZS36070) | |
| | 40 | **ZS1264I0** | ENCODER KNOB | | エンコーダーツマミ | | |
| | 50 | **ZP0317I0** | VOLUME KNOB | | ボリュームツマミ | | |
| | 70 | **ZP435500** | BATTERY COVER | | 電池蓋Ass'y | | |
| | 80 | **WE98740R** | BIND HEAD TAPPING SCREW-B | 3.0X12 MFZN2W3 | Bタイト+BIND | | 8 |
| | 90 | **ZE329300** | PW HEAD TAPPING SCREW-B | 3.0X30 MFZN2W3 | Bタイト+PWH | | 3 |
| | 100 | **WF491002** | BIND HEAD TAPPING SCREW-B | 3.0X30 MFZN2W3 | Bタイト+BIND | | 4 |
| | 110 | **WE774302** | BIND HEAD TAPPING SCREW-B | 3.0X8 MFZN2W3 | Bタイト+BIND | | 2 |
| | 120 | **ZS1265I0** | CONTROL KNOB | | コントロールノブ | | 2 |
| | 130 | **ZS1260I0** | LCD PANEL MOLDING | | LCDパネル成形品 | | |
| | 140 | **ZS438400** | EMBLEM | | エンブレム | | |
| | | | | | | | |
| | | | ACCESSORIES | | 付属品 | PSR-E463 | |
| | | **ZP509400** | MUSIC REST WITH BAG | BLACK | 譜面板袋入り | | |
| | | **ZT018300** | AC ADAPTOR SET | PA-150B E YES | アダプターセット | E | |
| | | **ZT041400** | AC ADAPTOR SET | PA-150B B YES | アダプターセット | B | |
| | | **WK014701** | AC ADOPTOR | PA-150U U | ACアダプター | C | |
| | | **ZT041500** | AC ADAPTOR SET | PA-150B K YES | アダプターセット | K | |
| | | **ZT041600** | AC ADAPTOR SET | PA-150B BRA YES | アダプターセット | P | |
| | | **ZT041900** | AC ADAPTOR SET | PA-150B A YES | アダプターセット | A | |
| * | | **ZU900500** | AC ADAPTOR SET | PA-150B IN YES | アダプターセット | Z | |
| | | **ZT041700** | AC ADAPTOR SET | PA-150B J YES | アダプターセット | J | |
| | | **ZT041800** | AC ADAPTOR SET | PA-150B CHN YES | アダプターセット | O | |
| * | | **ZZ258000** | JAPANESE SHEET SET | PK PSR-E463 | 和文シート袋入り | J | |
| * | | **ZZ258100** | CHINESE SHEET SET | PK PSR-E463 | 中文シート袋入り | O | |

＊: New Parts（新規部品）

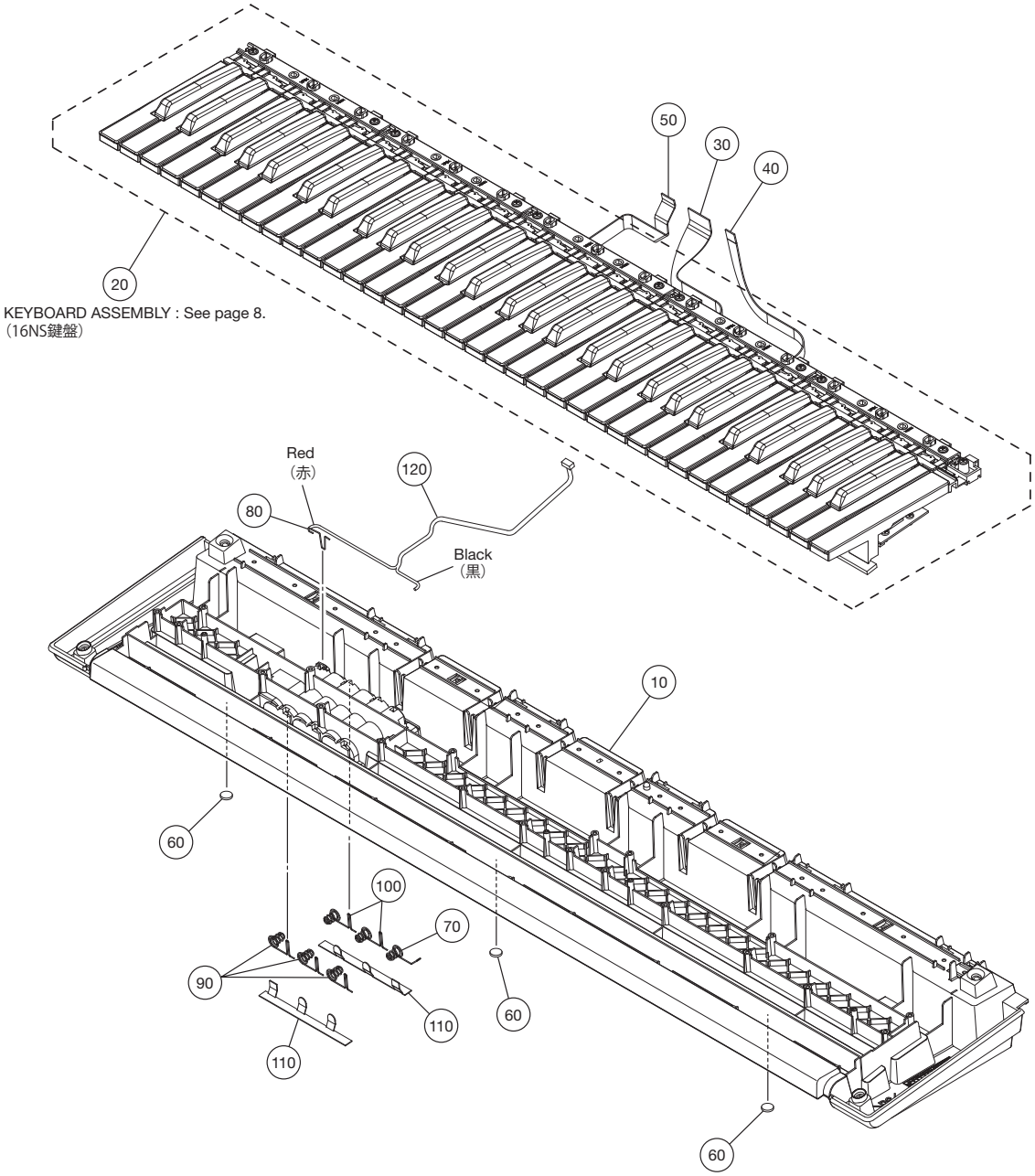
## ■ UPPER CASE ASSEMBLY（上ケース Ass'y）



| | REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|---|
| | | -- | UPPER CASE ASSEMBLY | PK PSR-E463 | 上ケースAss'y | (ZZ25730) | |
| * | 10 | ZZ132900 | UPPER CASE FINISHED | PSR-E463 | 上ケース塗装印刷品 | | |
| | 20 | ZS100600 | SP GRILLE (R) | BLACK | SPグリル(R) | | |
| | 30 | ZS100400 | SP GRILLE (L) | BLACK | SPグリル(L) | | |
| | 50 | ZR960600 | PN SWITCH POWER | RUBBER CONTACT X1 | PNスイッチパワー | | |
| | 60 | ZR960700 | PN SWITCH DEMO | RUBBER CONTACT X8 | PNスイッチデモ | | |
| | 70 | ZR960800 | PN SWITCH VOICE | RUBBER CONTACT X4 | PNスイッチボイス | | |
| | 80 | ZR960900 | PN SWITCH FUNC | RUBBER CONTACT X2 | PNスイッチファンク | | 2 |
| | 90 | ZR961000 | PN SWITCH TENKEY | RUBBER CONTACT X12 | PNスイッチテンキー | | |
| * | 100 | ZZ121100 | PN SWITCH SEC | RUBBER CONTACT X6 | PNスイッチセク | | 2 |
| * | 110 | ZZ121200 | PN SWITCH REGIST | RUBBER CONTACT X5 | PNスイッチレジスト | | |
| | 120 | ZR961400 | PN SWITCH DJ | RUBBER CONTACT X4 | PNスイッチDJ | | |
| | 130 | ZR961500 | PN SWITCH ASSIGN | RUBBER CONTACT X1 | PNスイッチアサイン | | 2 |
| * | 200 | ZY337900 | CIRCUIT BOARD | DMLCD(H) | DMLCD(H)シート | | |
| * | 210 | VAK55900 | CIRCUIT BOARD | PNL-T (AMPN0) | PNL-TシートSP | | |
| | 220 | ZV086300 | CIRCUIT BOARD | PNL-B(PN0) | PNL-BシートSP | | |
| * | 230 | ZZ263700 | CIRCUIT BOARD | PNR-T (AMPN1) | PNR-Tシート | | |
| * | 240 | ZZ263800 | CIRCUIT BOARD | ENC (AMPN2) | ENCシート | | |
| * | 250 | ZZ263900 | CIRCUIT BOARD | DC-IN EH (AMPN3) | DC-IN EHシート | | |
| | 260 | ZT314400 | CIRCUIT BOARD | PNR-B(PN1) | PNR-Bシート | | |
| | 270 | ZT314500 | CIRCUIT BOARD | KNOB(PN2) | KNOBシート | | |
| | 280 | ZT314600 | CIRCUIT BOARD | MVR(PN3) | MVRシート | | |
| | 290 | ZT314700 | CIRCUIT BOARD | PB(PN4) | PBシート | | |
| * | 302 | ZZ092000 | BACK LIGHT ASSEMBLY | | バックライトAss'y | | |
| * | 304 | ZZ017200 | LCD DISPLAY | YDDC0119AAHDTZS | 液晶ディスプレイ | | |
| | 306 | WZ963200 | RUBBER CNNECTOR | ZEBRA | ゴムコネクター | | 2 |
| | 310 | ZT045000 | WHEEL ASSEMBLY | | ホイールAss'y | | |
| | 330 | VE968501 | GRISE | G-31KA(KANTO-KASEI | グリス | | |
| | 380 | WE774302 | BIND HEAD TAPPING SCREW-B | 3.0X8 MFZN2W3 | Bタイト+BIND | | 69 |
| | 440 | -- | CUSHION PU | 123X5X5 | クッションPU | (ZT31720) | 2 |
| | 460 | -- | CUSHION PE | 115X13X1 | クッションPE | (ZT31730) | |
| | 470 | ZT045300 | DUST PROOF CUSHION | 123X4X2 | 防塵クッション | | 2 |
| | 480 | ZT045400 | DUST PROOF CUSHION | 184X4X1 | 防塵クッション | | |
| | 530 | -- | CONNECTOR ASSEMBLY | PNL PH-PH 7PL=135 | PNL束線 | (ZT31680) | |
| | 540 | -- | CONNECTOR ASSEMBLY | KNOB PH-PH 11P L=220 | KNOB束線 | (ZT31690) | |
| | 610 | ZT315400 | LIGHT GUIDE | | ライトガイド | | 3 |

＊: New Parts (新規部品)

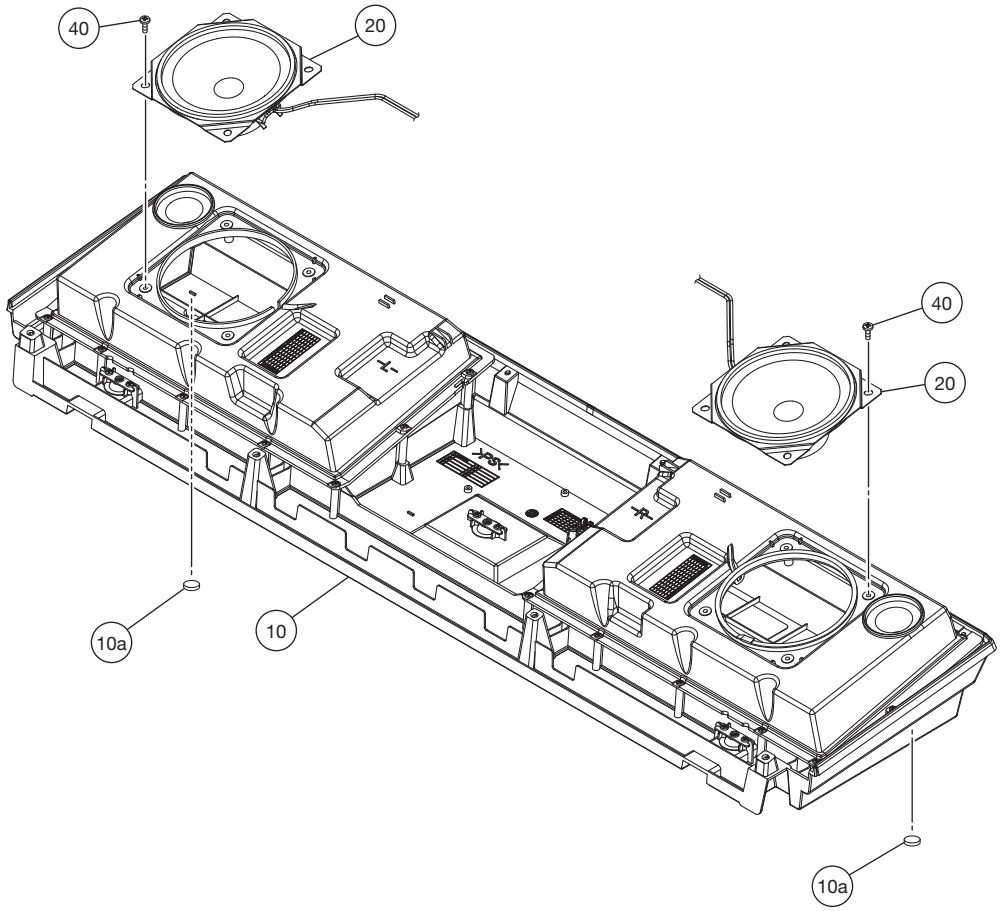
# ■ LOWER KEY BED ASSEMBLY（下ケース鍵盤 Ass'y）




| REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|
| | -- | LOWER KEY BED ASSEMBLY | | 下ケース鍵盤Ａｓｓ'ｙ | (ZS36080) | |
| 10 | **ZS1257I0** | LOWER CASE F MOLDING | | 下ケース成形品Ｆ | | |
| 20 | -- | 16NS-C61 KEYBOARD | 16NS 61 2M | １６ＮＳ－Ｃ６１鍵盤 | (ZT09950) | |
| 30 | -- | WIRING ASSEMBLY | MKH2 12P L=260 P=2.0 | ＭＫＨ２束線 | (ZT31590) | |
| 40 | -- | WIRING ASSEMBLY | MKH1 5P L=290 P=2.0 | ＭＫＨ１束線 | (ZT31580) | |
| 50 | -- | WIRING ASSEMBLY | MKL 7P L=220 P=2.0 | ＭＫＬ束線 | (ZT31570) | |
| 60 | **WW693500** | RUBBER FOOT | | ゴム脚 | | 3 |
| 70 | **WD87920R** | BATTERY TERMINAL A | | 接点バネＡ | | |
| 80 | **WD87930R** | BATTERY TERMINAL B | | 接点バネＢ | | |
| 90 | **ZU223500** | BATTERY TERMINAL | | 接点バネＣ改 | | 3 |
| 100 | **ZU223700** | BATTERY TERMINAL | | 接点バネＤ改 | | 2 |
| 110 | **ZP656600** | NONWOVEN CLOTH | 98X20X0.7T | 不織布 | | 2 |
| 120 | -- | CONNECTOR ASSEMBLY | BATT XH3P L=315 | ＢＡＴＴ束線 | (ZT31700) | |

＊: New Parts（新規部品）

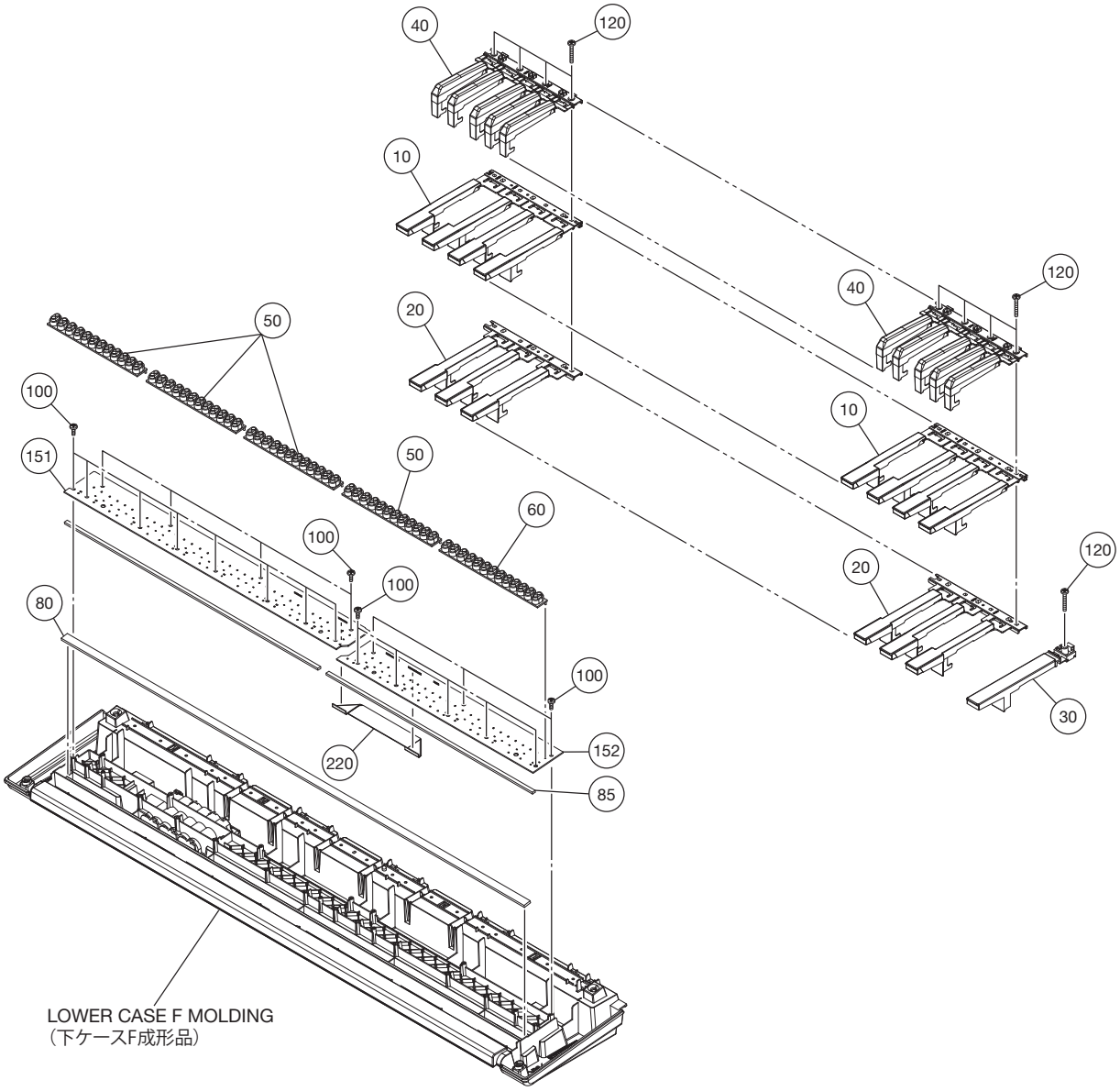
## ■ LOWER CACE ASSEMBLY（下ケース Ass'y）




| REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|
| | -- | LOWER CASE ASSEMBLY | | 下ケースAss'y | (ZS36070) | |
| 10 | ZT045900 | LOWER CASE SUB ASSEMBLY | | 下ケースサブAss'y | | |
| 10a | WW693500 | RUBBER FOOT | | ゴム脚 | | 2 |
| 20 | YH583A00 | LOUD SPEAKER ASSEMBLY | 12.0CM 6OHM 6W | スピーカーAss'y | | |
| 40 | WE97460R | BIND HEAD TAPPING SCREW-B | 4.0X8 MFZN2W3 | Bタイト＋BIND | | 8 |

＊: New Parts（新規部品）

# ■ KEYBOARD ASSEMBLY（16NS 鍵盤）




| REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|
| | -- | 16NS-C61 KEYBOARD | 16NS 61 2M | １６ＮＳ－Ｃ６１鍵盤 | (ZT09950) | |
| 10 | **ZE3698I0** | WHITE KEY | 16NS CEGB | 白鍵　ＣＥＧＢ | | 5 |
| 20 | **ZE3699I0** | WHITE KEY | 16NS DFA | 白鍵　ＤＦＡ | | 5 |
| 30 | **V47603I0** | WHITE KEY 16N | | 白鍵　Ｃ | | |
| 40 | **WZ1683I0** | BLACK KEY T=1.5 | 16NS # | 黒鍵 | | 5 |
| 50 | **ZS089000** | KEYBOARD RUBBER CONT | 16N-2M OCT 2M | 接点ゴム　１６Ｎ２Ｍ | | 4 |
| 60 | **ZS089100** | KEYBOARD RUBBER CONT | 16N-2M 13 2M | 接点ゴム　１６Ｎ２Ｍ | | |
| 80 | **VZ303002** | FELT L | 827 11 SHIRO | フェルト　Ｌ | | |
| 85 | **VZ302902** | FELT U | 836 5 SHIRO | フェルト　Ｕ | | |
| 100 | **WE774302** | BIND HEAD TAPPING SCREW-B | 3.0X8 MFZN2W3 | Ｂタイト＋ＢＩＮＤ | | 20 |
| 120 | **WF49200R** | BIND HEAD TAPPING SCREW-P | 3.0X20 MFZN2W3 | Ｐタイト＋ＢＩＮＤ | | 11 |
| 151 | **ZS616600** | CIRCUIT BOARD | MKL 16NS-61-2M-L | ＰＣＢ　ＭＫＬ（Ｈ） | | |
| 152 | **ZS617000** | CIRCUIT BOARD | MKH 16NS-61-2M-H | ＰＣＢ　ＭＫＨ（Ｈ） | | |
| 220 | -- | WIRING ASSEMBLY | MK-MK 13P L=194 P=2.0 | ＭＫ－ＭＫ束線 | (ZV57500) | |

＊: New Parts（新規部品）

# ■ ELECTRICAL PARTS（電気部品）

## DMLCD, PNL

| | REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|---|
| * | | **ZY337900** | CIRCUIT BOARD | DMLCD(H) PSR-E463 | ＤＭＬＣＤ（Ｈ）シート | (YJ549C0) | |
| * | | **VAK55900** | CIRCUIT BOARD | PNL-T (AMPN0) | ＰＮＬ－Ｔシート ＳＰ | (ZZ26360)(YH381E0) | |
| * | | **ZZ263700** | CIRCUIT BOARD | PNR-T (AMPN1) | ＰＮＲ－Ｔシート | (ZZ26360)(YH381E0) | |
| * | | **ZZ263800** | CIRCUIT BOARD | ENC (AMPN2) | ＥＮＣシート | (ZZ26360)(YH381E0) | |
| * | | **ZZ263900** | CIRCUIT BOARD | DC-IN EH (AMPN3) | ＤＣ－ＩＮ ＥＨシート | (ZZ26360)(YH381E0) | |
| | | **ZV086300** | CIRCUIT BOARD | PNL-B(PN0) | ＰＮＬ－Ｂシート ＳＰ | (ZS31790)(YH382C0) | |
| | | **ZT314400** | CIRCUIT BOARD | PNR-B(PN1) | ＰＮＲ－Ｂシート | (ZS31790)(YH382C0) | |
| | | **ZT314500** | CIRCUIT BOARD | KNOB(PN2) | ＫＮＯＢシート | (ZS31790)(YH382C0) | |
| | | **ZT314600** | CIRCUIT BOARD | MVR(PN3) | ＭＶＲシート | (ZS31790)(YH382C0) | |
| | | **ZT314700** | CIRCUIT BOARD | PB(PN4) | ＰＢシート | (ZS31790)(YH382C0) | |
| | | **ZS616600** | CIRCUIT BOARD | MKL 16NS-C61-2M-L | ＭＫＬシート | (YH445C0) | |
| | | **ZS617000** | CIRCUIT BOARD | MKH 16NS-C61-2M-H | ＭＫＨシート | (YH444C0) | |
| | | | | | | | |
| * | | **ZY337900** | CIRCUIT BOARD | DMLCD(H) PSR-E463 | ＤＭＬＣＤ（Ｈ）シート | (YJ549C0) | |
| | CN401 | **WH382500** | CONNECTOR | UAR27 4P SE | ＵＳＢ コネクタ | USB TO DEVICE | |
| | CN401 | **--** | CONNECTOR | YKF45-0033N 4P SE | ＵＳＢ コネクタ | USB TO DEVICE (WK45070) | |
| | CN451 | **V680260R** | CONNECTOR | USB 4P SE | ＵＳＢ ジャック | USB TO HOST | |
| | CN451 | **--** | CONNECTOR | YKF45-0021N 4P SE | ＵＳＢ コネクタ Ｂ | USB TO HOST (V901600) | |
| | JK201 | **VV943300** | CONNECTOR | JACK HTJ-064-04A | ホーンコネクタ | PHONES/OUTPUT | |
| | JK201 | **--** | CONNECTOR | MSJ-064-15A B AG | ホーンコネクタ | PHONES/OUTPUT (WJ30620) | |
| | JK301 | **VC68750R** | CONNECTOR | JACK YKB21-5014 | ホーンコネクタ（黒） | SUSTAIN | |
| | JK701 | **--** | CONNECTOR | JACK YKB21-5352N | ホーンコネクタ | AUX-IN (VB23040) | |
| | JK701 | **ZA590001** | CONNECTOR | JACK MINI STEREO | ホーンコネクタ | AUX-IN | |
| | C212 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C213 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C220 | **UF038103** | CAPACITOR | 100 16V | チップケミコン | | |
| | C221 | **UF038103** | CAPACITOR | 100 16V | チップケミコン | | |
| | C228 | **UF14810R** | CAPACITOR | 100 25V | チップケミコン | | |
| | C230 | **UF14810R** | CAPACITOR | 100 25V | チップケミコン | | |
| | C276 | **ZC041800** | ELECTROLYTIC CAPACITOR | 220.00 25.0V | チップケミコン ＣＬ | | |
| | C402 | **UF038103** | CAPACITOR | 100 16V | チップケミコン | | |
| | C605 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C706 | **UF03722R** | CAPACITOR | 22 16V | チップケミコン | | |
| | C707 | **UF03722R** | CAPACITOR | 22 16V | チップケミコン | | |
| | C714 | **UF03722R** | CAPACITOR | 22 16V | チップケミコン | | |
| | C715 | **UF03722R** | CAPACITOR | 22 16V | チップケミコン | | |
| | C718 | **UF038103** | CAPACITOR | 100 16V | チップケミコン | | |
| | C723 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C725 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C727 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C729 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | C802 | **UF05710R** | CAPACITOR | 10 35V | チップケミコン | | |
| | C803 | **UF14810R** | CAPACITOR | 100 25V | チップケミコン | | |
| | C812 | **UF038103** | CAPACITOR | 100 16V | チップケミコン | | |
| | C815 | **UF037101** | CAPACITOR | 10 16V | チップケミコン | | |
| | IC001 | **YG320A00** | IC | MB8AA4640 | ＣＰＵ／周辺ＩＣ | | |
| * | IC101 | **YJ745B00** | IC | WAVE/PROGRAM 00084 | 書込済ＲＯＭ | | |
| | IC111 | **X9347A01** | IC | R3112N291A-TR-FE | 電源検出ＩＣ | | |
| * | IC121 | **YC145B00** | IC | MX29GL128FLT2I-90G | メモリＩＣ | | |
| | IC131 | **XZ414G00** | IC | W9864G6KH-6 SDRAM | メモリＩＣ | | |
| | IC201 | **YH390A00** | IC | AK4430 | ＩＣ ＤＡＣ | | |
| | IC211 | **YH100A02** | IC | NJM8080G(TE2) | アンプＩＣ ＳＯＰ | | |
| | IC251 | **YF663B00** | IC | YDA176-QZE2 | ＩＣ | | |
| | IC401 | **YD235A00** | IC | R5524N002A-TR-FE | 電源ＩＣ | | |
| | IC401 | **--** | IC | BD2232G-GTR | 電源ＩＣ | (YF575A0) | |
| | IC601 | **XZ987A03** | IC | ML9040A-B01GAZ03A | ＣＰＵ／周辺ＩＣ | | |
| | IC701 | **YH100A02** | IC | NJM8080G(TE2) | アンプＩＣ ＳＯＰ | | |
| | IC702 | **X7357B00** | IC | PCM1803ADBR | ＩＣ | | |
| | IC801 | **YE332A00** | IC | R1245K003A-TR | 電源ＩＣ | | |
| | IC802 | **YD113A00** | IC | RP131H331D-T1-FE 3 | 電源ＩＣ | | |
| | R222 | **RD15433R** | CARBON RESISTOR (CHIP) | 33.0 1/4 J TP | チップ抵抗 | | |
| | R223 | **RD15433R** | CARBON RESISTOR (CHIP) | 33.0 1/4 J TP | チップ抵抗 | | |
| | X101 | **WE19440R** | RESONATOR QUARTZ | 16.9344M HC-49S-SM | 水晶振動子 | | |
| | X102 | **ZJ113600** | RESONATOR QUARTZ | 48.000000M DSX321G | 水晶振動子 | | |
| | | | | | | | |
| * | | **VAK55900** | CIRCUIT BOARD | PNL-T (AMPN0) | ＰＮＬ－Ｔシート ＳＰ | (ZZ26360)(YH381E0) | |
| * | | **ZZ263700** | CIRCUIT BOARD | PNR-T (AMPN1) | ＰＮＲ－Ｔシート | (ZZ26360)(YH381E0) | |
| * | | **ZZ263800** | CIRCUIT BOARD | ENC (AMPN2) | ＥＮＣシート | (ZZ26360)(YH381E0) | |
| * | | **ZZ263900** | CIRCUIT BOARD | DC-IN EH (AMPN3) | ＤＣ－ＩＮ ＥＨシート | (ZZ26360)(YH381E0) | |
| | EC351 | **VU48130R** | SWITCH OTHERS | REB161(9X5)PVB15FH | １６形エンコーダ | CATEGORY | |

＊: New Parts（新規部品）

# PNL, MKL

| | REF NO. | PART NO. | DESCRIPTION | | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|---|
| | JK101 | **WZ704401** | CONNECTOR | DC-502-AG-PBT-2.0- | 電源コネクタ | DC-IN | |
| | TR101 | **ZC633500** | TRANSISTOR | 2SB1568 | トランジスタ 2SB | | |
| | TR101 | **--** | TRANSISTOR | TTB1020B,S4X(S | トランジスタ 2SB | (ZW58930) | |
| | WH010 | **--** | WIRING ASSEMBLY | 5P 180MM P=2.0 | Fケーブル105 | (MF80518) | |
| ⚠ | WH020 | **--** | WIRING ASSEMBLY | 10P 140MM P=2.0 | Fケーブル105 | (MF81014) | |
| | WH030 | **--** | WIRING ASSEMBLY | 8P 180MM P=2.0 | Fケーブル105 | (MF80818) | |
| | WH040 | **--** | WIRING ASSEMBLY | 4P 100MM P=2.0 | Fケーブル105 | (MF80410) | |
| | C103 | **V8689300** | SEMICONDUCTOR CERAMIC CAP. | 0.1000 25V Z RX TP | 半導体セラコン | | |
| | C104 | **V8689300** | SEMICONDUCTOR CERAMIC CAP. | 0.1000 25V Z RX TP | 半導体セラコン | | |
| | C105 | **UR849100** | ELECTROLYTIC CAPACITOR | 1000 25.0V RX TP | ケミコン | | |
| | C106 | **FG64410R** | CERAMIC CAPACITOR | 0.0100 50V Z RX TP | セラコン(F) | | |
| | FZ101 | **WN081300** | FUSE | SEMKO/UL/PSE 2.00A | ヒューズ250V | | |
| | FZ101 | **--** | FUSE | SEMKO/UL/PSE 2.00A | ヒューズ250V | (ZU45890) | |
| | TR102 | **V279770R** | TRANSISTOR | 2SC5395-T112-E/F | トランジスタ 2SC | | |
| | TR102 | **--** | TRANSISTOR | 2N5551C-AT/P | トランジスタ | (WC39840) | |
| ⚠ | D101 | **WD00970R** | DIODE | 10EDB40-TA1B2 TE-2 | ダイオード | | |
| ⚠ | D102 | **WD00970R** | DIODE | 10EDB40-TA1B2 TE-2 | ダイオード | | |
| | D103 | **ZQ176200** | DIODE | RL202 | ダイオード | | |
| | D105 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | D106 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | D204 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | -207 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | D253 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | -256 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | R101 | **HF457470** | CARBON RESISTOR | 47.0K 1/4 J AX TP | カーボン抵抗 | | |
| | R102 | **HF456330** | CARBON RESISTOR | 3.3K 1/4 J AX TP | カーボン抵抗 | | |
| | R103 | **HF45715R** | CARBON RESISTOR | 15.0K 1/4 J AX TP | カーボン抵抗 | | |
| | R104 | **HF457470** | CARBON RESISTOR | 47.0K 1/4 J AX TP | カーボン抵抗 | | |
| | | | | | | | |
| | | **ZV086300** | CIRCUIT BOARD | PNL-B(PN0) | PNL-Bシート SP | (ZS31790)(YH382C0) | |
| | | **ZT314400** | CIRCUIT BOARD | PNR-B(PN1) | PNR-Bシート | (ZS31790)(YH382C0) | |
| | | **ZT314500** | CIRCUIT BOARD | KNOB(PN2) | KNOBシート | (ZS31790)(YH382C0) | |
| | | **ZT314600** | CIRCUIT BOARD | MVR(PN3) | MVRシート | (ZS31790)(YH382C0) | |
| | | **ZT314700** | CIRCUIT BOARD | PB(PN4) | PBシート | (ZS31790)(YH382C0) | |
| | SW263 | **WG31840R** | SWITCH PUSH | SKRGAMD010 | タクト SW | KEYBOARD/BACKING | |
| | SW273 | **WG31840R** | SWITCH PUSH | SKRGAMD010 | タクト SW | KNOB ASSIGN | |
| | VR101 | **VQ67050R** | VR ROTARY | B 10K RK11K1130A0M | ロータリーボリューム | MASTER VOLUME | |
| * | VR301 | **ZV411300** | VR ROTARY | B 10.0K RK11K1130D | ロータリー VR | LIVE CONTROL A | |
| * | VR302 | **ZV411300** | VR ROTARY | B 10.0K RK11K1130D | ロータリー VR | LIVE CONTROL B | |
| | VR303 | **WZ510001** | VR ROTARY | B 10.0K RK11K1110D | ロータリー VR | PITCH BEND | |
| | WH010 | **--** | WIRING ASSEMBLY | 13P 80MM P=2.0 | Fケーブル | (MF81308) | |
| | WH020 | **--** | WIRING ASSEMBLY | 8P 80MM P=2.0 | Fケーブル | (MF80808) | |
| | WH030 | **--** | WIRING ASSEMBLY | 12P 80MM P=2.0 | Fケーブル | (MF81208) | |
| | WH040 | **--** | WIRING ASSEMBLY | 3P 160MM P=2.0 | Fケーブル | (MF80316) | |
| | WH050 | **--** | WIRING ASSEMBLY | 3P 100MM P=2.0 | Fケーブル | (MF80310) | |
| | LD200 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | GROOVE CREATOR | |
| | LD201 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | SONG/AUDIO | |
| | LD202 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | STYLE | |
| | LD203 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | GROOVE CREATOR | |
| | LD204 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | SONG | |
| | LD205 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | STYLE | |
| | LD206 | **ZT819900** | LED | CLM1B-RKW-CUBVBAA3 | チップLED | VOICE | |
| | D200 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | -203 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | D250 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | -252 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | D301 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | D302 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | R200 | **HF455180** | CARBON RESISTOR | 180.0 1/4 J AX TP | カーボン抵抗 | | |
| | -202 | **HF455180** | CARBON RESISTOR | 180.0 1/4 J AX TP | カーボン抵抗 | | |
| | R203 | **HF45468R** | CARBON RESISTOR | 68.0 1/4 J AX TP | カーボン抵抗 | | |
| | -206 | **HF45468R** | CARBON RESISTOR | 68.0 1/4 J AX TP | カーボン抵抗 | | |
| | | | | | | | |
| | | **ZS616600** | CIRCUIT BOARD | MKL 16NS-C61-2M-L | MKLシート | (YH445C0) | |
| * | D1 | **VB941201** | DIODE | 1SS133,1SS176 TE-5 | ダイオード | | |
| * | -72 | **VB941201** | DIODE | 1SS133,1SS176 TE-5 | ダイオード | | |
| | D1 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | -72 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | | |
| | | | | | | | |
| | | **ZS617000** | CIRCUIT BOARD | MKH 16NS-C61-2M-H | MKHシート | (YH444C0) | |

＊: New Parts（新規部品）

## MKH

| | REF NO. | PART NO. | DESCRIPTION | 部品名 | REMARKS | QTY |
|---|---|---|---|---|---|---|
| * | D73 | **VB941201** | DIODE | 1SS133,1SS176 TE-5 | ダイオード | |
| * | -122 | **VB941201** | DIODE | 1SS133,1SS176 TE-5 | ダイオード | |
| | D73 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | |
| | -122 | **VD631601** | DIODE | 1SS133,176,HSS104 | ダイオード | |

＊: New Parts（新規部品）

# PSR-E463 BLOCK DIAGRAM

TOUCH RESPONSE
Keyboard 61KEY (16N)

MKL (C1~B3) — CN4(12P), CN3(12P) — SW1-36 — CN5(7P)

MKH (C4~C6) — SW37-61 — CN2(5P), CN1(12P)

ENC — ENCODER, EC351 — EN/A, EN/B — CN351(4P)

PNR-B — SW 204-206,214-216,224-226 — SW MATRIX ( SW9 PCS ) — PPB[4-6] — LD203-206 LED (4PCS) — PPA[3-6], PPA[0-2] — PPB[4-7], PPA[3-6] — LED[1] — PPB[4-7], PPA[0-6] — CN252(8P) — CN251(12P)

PNR-T — SW 234-237,244-247, 254-257,264-267 — SW MATRIX ( SW16 PCS ) — PPB[4-7] — PPA[3-6] — CN151(8P)

KNOB — Knob A, Knob B — VR301, VR302 — DSP SW273 — ASSIGN SW263 — PPA[6], PPA[7], PPB[3] — Knob_A centerA, Knob_B centerB, PB — CN301(11P) — CN302(3P)

PB — VR303, PB — CN303(3P)

CN501(7P), CN502(5P), CN503(12P), CN304(4P), CN302(12P), CN301(13P)

## DMLCD

B[01-07], B[07-11], N[11-16],N[21-26]

IC501 ECORDER 74LV138A

IC502 DECORDER 74LV138A

+5D — H · S · SW R5524N002A — IC401 — USB1.1 Full speed

To Device — CN401

To Host — CN451

Sustain — JK301

DP0, DM0 POW_ENB,/OVER_C

/DAMP-MUTE — PA0

/DETECT-CHECK — PA1

/MUTE — PA2

FUNC_DP, FUNC_DM VBUS, PULLUP_ENB — USB1.1 Full speed

AUXIN-IN — PA35

SUSTAIN — PC7

RESET +2.9V 44.9ms — IC111 — /RESET — RESN

HOST_DP HOST_DM /OVER_C POWER_ENB — KYB01-04 — PB2 — PB14 — KYN11-16 — KYN21-26 — PA32-33 — PB1 — PA24-31 — PA3-12

HeartBeat, SP_OFF, EN/A,EN/B, LED-OUT

PB8 — /IC2 — IC351 — KnobB, centerB, KnobA, centerA, PB, MVR — VR_A,VR_B, PB_MVR

AN0-2, AN3 — Battery Monitor

/PSWI, PSWO

XI, XO — RESONATOR QUARTZ 16.9344MHz F Tolerance±50PPM — X101

UCLK_I, UCLK_O — RESONATOR QUARTZ 48MHz F Tolerance±50PPM — X102

CPU SWX03 IC001

MA01-24, MD00-15, WEN,RDN, LDQM,UDQM, RCLK, CSDN,1N,2N, PB16, SDI,SDO0,SDO1, SYSCK, BCLK,WCLK

/CS1,/WE, LDQM,UDQM,RCLK, MA[1-12],[14-15], /RAS(MA[17]), /CAS(MA[18]), MD[0-15]

/CS2, /WE,/RD, MA[1-23], MD[0-15]

/CS0, /WE,/RD, MA[1-24], MD[0-15]

SDRAM 64M IC131

FLASH ROM 128M IC121

PROG/WAVE ROM 256M IC101

/ADC_IC — ADC 24bit PCM1803A — IC702

SDI,SYSCLK BCLK,WCLK — DAC 24bit AK4430 — IC201 — SDO1,SYSCLK BCLK,WCLK

SDO0,SYSCLK BCLK,WCLK — /HP-MUTE — /DAMP_MUTE — D.AMP YDA176 IC251 — PVDD

RC Filter — /HP-MUTE — MUTE Circuit — TR203-206 — NO MOUNT — MUTE Circuit

LC Filter — CN251(4P) — Rch, Lch — SPEAKERS PSR-E463: 6Ω/12cm

AUX IN — JK701 — AUXIN-IN

PHONES — JK201 — HP-IN

TR305 LED DRIVER — LED[1], PPB[4-7], PPA[0-6]

TR301-304 LED DRIVER — PA[0-7]

IC601 LCD DRIVER — PA[0-7] RS,ENB

CN305(2P)

COM[1-16] SEG[1-40] — LCD DISPLAY LD601 — BACK LIGHT LED:1PC(Blue) — LCD

HeartBeat — Error Detection IC151 — /MUTE, HP-IN, SP_OFF — TR151, TR152 — /DETECT — IC152(1/4) — /HP-MUTE, /DETECT-CHECK — IC152(2/4), IC152(3/4) — /DAMP-MUTE

TR202 Ripple Filter — +HP, +5A, +5D, +3.3A, +3.3D — +3.3V REGULATOR IC802 — +5V DC/DC IC801 — PVDD, +B

PB Knob_A, centerA Knob_B, centerB

CN303(8P)

CN201(13P), CN204(11P)

PPB[0-3], LED[1], PPA[0-2] — LED (3PCS) LD200-202

SW200-202 SW210-212 SW220-222 SW230-232 — SW MATRIX ( SW12 PCS ) — PPB[0-2]

PPA[6-7], PPB[3], PPA[0-3]

PB Knob_A, centerA Knob_B, centerB

CN202(8P), CN203(7P) — PPB[0-2], PPA[4-7]

## PNL-B

CN106(7P)

## PNL-T

PPB[0-2]

SW 240-242,250-252, 260-262,270-272 — SW MATRIX ( SW12 PCS ) — PPA[4-7]

## MVR

MASTER VOLUME VR101 — 10KΩB — CN107(3P)

CN105(3P), CN104(10P)

+3.3D, MASTER-VR

+B, /PSWI, PSWO — TR101, TR102 — FZ101

STANDBY/ON * More than 400ms SW101

CN801(10P)

MVR, +B, /PSWI, PSWO

CN102(5P), CN101(5P)

## DC-IN

DC-IN — JK101

AC adaptor PSR-E463 PA-150B or equivalent

CN103(3P)

Battery 1.5×6 — "AA"size,LR6 or equivalent batteries

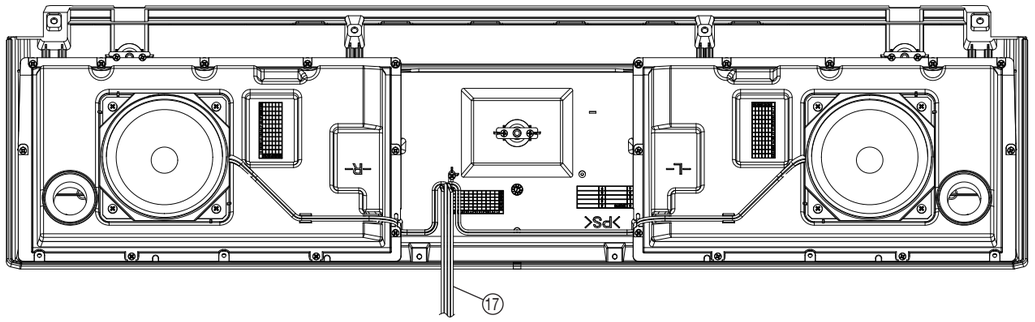
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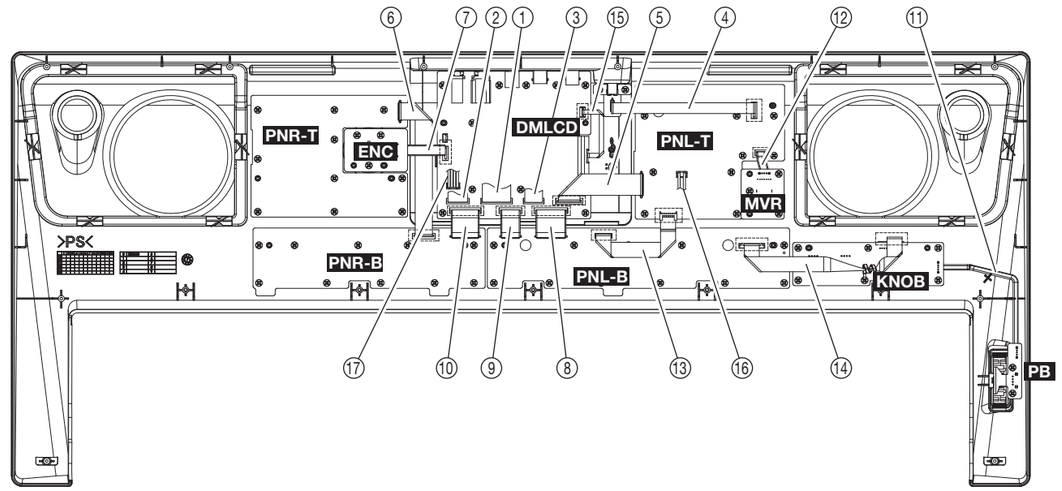
PSR-E463

## ■ PSR-E463 CIRCUIT BOARD LAYOUT & WIRING

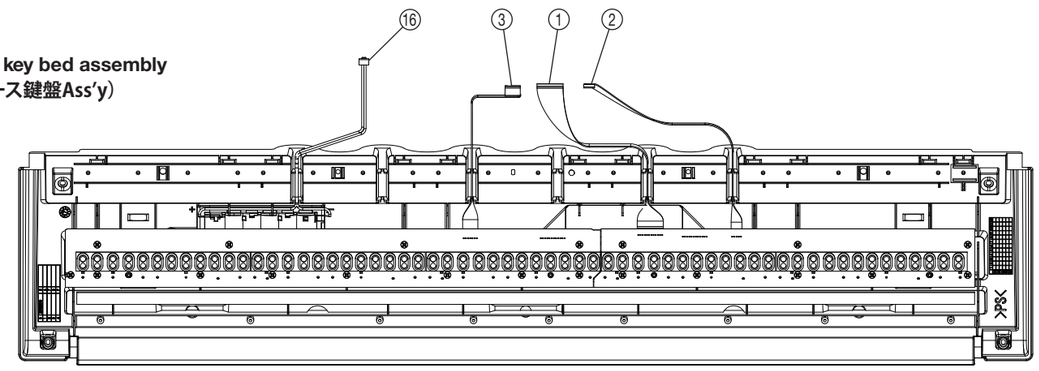
● **Lower case assembly**
**（下ケースAss'y）**




● **Upper case assembly**
**（上ケースAss'y）**




● **Lower key bed assembly**
**（下ケース鍵盤Ass'y）**




| No. | Location（ロケーション） | Parts No.（部品番号） | Connector Assembly（束線名） | Destination（配線先） | | | | | | Remarks（備考） |
|---|---|---|---|---|---|---|---|---|---|---|
| ① | 30 | (ZT31590) | WIRING ASSEMBLY **MKH2** | MKH - CN01 | *1 | *4 | DMLCD - CN503 | *1 | *4 | 12P L=260 |
| ② | 40 | (ZT31580) | WIRING ASSEMBLY **MKH1** | MKH - CN02 | *1 | *4 | DMLCD - CN502 | *1 | *4 | 5P L=290 |
| ③ | 50 | (ZT31570) | WIRING ASSEMBLY **MKL** | MKL - CN05 | *1 | *4 | DMLCD - CN501 | *1 | *4 | 7P L=220 |
| ④ | WH010 | (MF80518) | WIRING ASSEMBLY | DC-IN - CN101 | *3 | *10 | PNL-T - CN102 | *1 | *4 | 5P L=180 |
| ⑤ | WH020 | (MF81014) | WIRING ASSEMBLY | PNL-T - CN104 | *3 | *10 | DMLCD - CN801 | *1 | *4 | 10P L=140 |
| ⑥ | WH030 | (MF80818) | WIRING ASSEMBLY | PNR-T - CN151 | *3 | *10 | PNR-B - CN252 | *1 | *4 | 8P L=180 |
| ⑦ | WH040 | (MF80410) | WIRING ASSEMBLY | ENC - CN351 | *3 | *10 | DMLCD - CN304 | *1 | *4 | 4P L=100 |
| ⑧ | WH010 | (MF81308) | WIRING ASSEMBLY | PNL-B - CN201 | *3 | *10 | DMLCD - CN301 | *1 | *4 | 13P L=80 |
| ⑨ | WH020 | (MF80808) | WIRING ASSEMBLY | PNL-B - CN202 | *3 | *10 | DMLCD - CN303 | *1 | *4 | 8P L=80 |
| ⑩ | WH030 | (MF81208) | WIRING ASSEMBLY | PNR-B- CN251 | *3 | *10 | DMLCD - CN302 | *1 | *4 | 12P L=80 |
| ⑪ | WH040 | (MF80316) | WIRING ASSEMBLY | KNOB - CN302 | *3 | *10 | PB - CN303 | *1 | *4 | 3P L=160 |
| ⑫ | WH050 | (MF80310) | WIRING ASSEMBLY | MVR - CN107 | *3 | *10 | PNL-T - CN105 | *1 | *4 | 3P L=100 |
| ⑬ | 530 | (ZT31680) | CONNECTOR ASSEMBLY **PNL** | PNL-B - CN203 | *1 | *8 | PNL-T - CN106 | *1 | *8 | 7P L=135 |
| ⑭ | 540 | (ZT31690) | CONNECTOR ASSEMBLY **KNOB** | PNL-B - CN204 | *1 | *8 | KNOB - CN301 | *1 | *8 | 11P L=220 |
| ⑮ | | | BACK LIGHT ASSEMBLY | BACK LIGHT ASSEMBLY | *11 | *10 | DMLCD - CN305 | *1 | *8 | 2P |
| ⑯ | 120 | (ZT31700) | CONNECTOR ASSEMBLY **BATT** | CONTACT SPRING (+) | *2 | *6 | PNL-T - CN103 | *1 | *8 | XH3P L=315 |
| | | | | CONTACT SPRING ( –) | *2 | *6 | | | | |
| ⑰ | | | SPEAKER ASSEMBLY | SPEAKER L | *11 | *10 | DMLCD - CN251 | *1 | *8 | XH4P |
| | | | | SPEAKER R | *11 | *10 | | | | |

* The parts with "( )" in "Part No." are not available as spare parts.
*1: Installation
*2: Manual soldering
*3: Dip soldering
*4: Edge mark is adjusted to Pin 1 mark ( △ mark).
*5: Edge mark is adjusted to + mark.
*6: Red wire is connected to (+) terminal. Black wire is connected to (Ð) terminal.
*7: White wire is connected to (+) terminal. Black wire is connected to (Ð) terminal.
*8: Red wire is adjusted to Pin 1 mark ( △ mark).
*9: White wire is adjusted to Pin 1 mark ( △ mark).
*10: Connected
*11: Connected as Ass'y

**Caution**: Be sure to attach the removed filament tape just as it was before removal.

* 部品番号が ( ) で囲まれている部品は、サービス部品として準備されていません。
*1: 差込み
*2: 手半田
*3: ディップ
*4: エッジマークが 1 ピン側 ( △ )
*5: エッジマークが + 側 (+)
*6: 赤色線材が (+) 端子、黒色線材が (–) 端子
*7: 白色線材が (+) 端子、黒色線材が (–) 端子
*8: 赤色線材が 1 ピン側 ( △ )
*9: 白色線材が 1 ピン側 ( △ )
*10: 接続済
*11: ASS'Yとして接続済

**注意：**一度剥がしたフィラメントテープは、取り外す前と同じように取り付けてください。


PSR-E463

# PSR-E463 OVERALL CIRCUIT DIAGRAM (DMLCD)

< Page 1 >

DMLCD

LCD DRIVER
IC401
ML9040A-B01GA

SDRAM 16M

FLASH ROM 16M

RESET

PROG/WAVE ROM 256M

DECORDER

CPU

HIGH SIDE SW

ERROR DETECTION

DAC

D.AMP

ADC

+3.3V REGULATOR

+5V DC/DC

Headphone Amplifier
Total Gain:3.69dB

OUTPUT Amplifie

NO MOUNT

to PNL-B-CN201
<Page 2 F5>

to PNR-B-CN251
<Page 2 B4>

to PNL-B-CN202
<Page 2 F6>

to ENC-CN351
<Page 3 B3>

LCD Backlight

SUSTAIN

to PNL-T-CN104
<Page 3 D7>

PHONES

SPEAKERS

PSR-E463:6Ω

to MKH-CN2
<Page 4 B4>

to MKL-CN5
<Page 4 B8>

to MKH-CN1
<Page 4 B2>

USB TO DEVICE

USB TO HOST

AUX-IN

| | R255 | R256 | R257 | R258 | Limit W | Gain Mode |
|---|---|---|---|---|---|---|
| PSR-E463 | No Mount | 0 | 68k | 10k | 6W | 2 |
| | M | | K | | | |

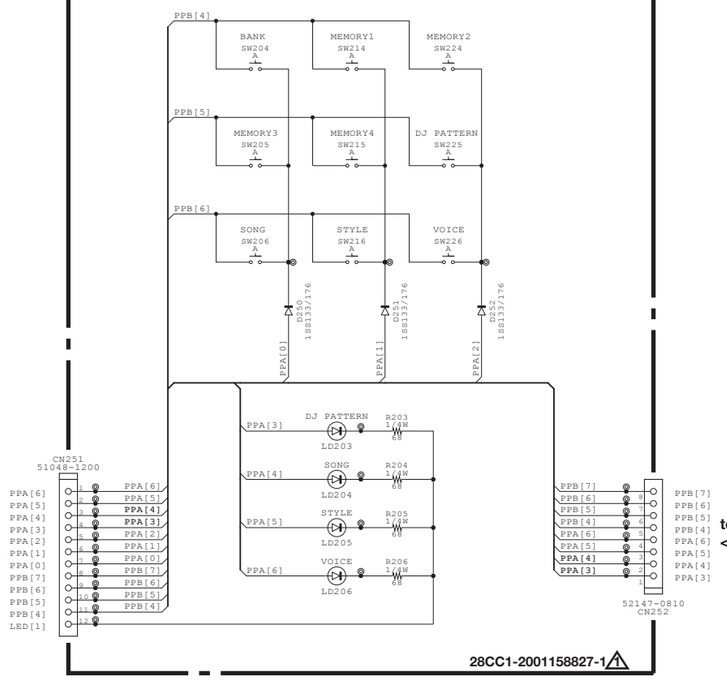
| | R801 | R802 | R11k |
|---|---|---|---|
| PSR-E463 | Mount | No Mount | SY33790 |

| | C277,C278,C279 |
|---|---|
| PSR-E463 | No Mount |

# ■ PSR-E463 OVERALL CIRCUIT DIAGRAM (PNR-B,PNL-B,MVR,PB,KNOB)

< Page 2 >

## PNR-B

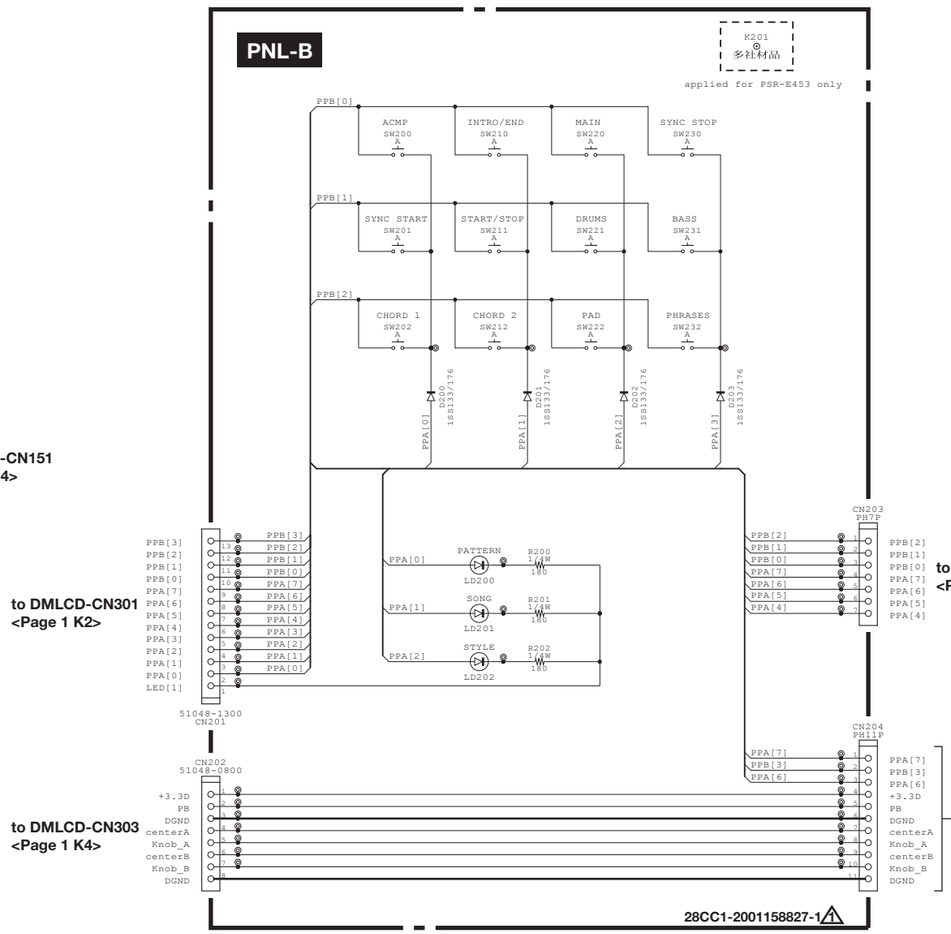



to DMLCD-CN302 <Page 1 K3>

to PNR-T-CN151 <Page 3 I4>

28CC1-2001158827-1 ⚠1

## PNL-B



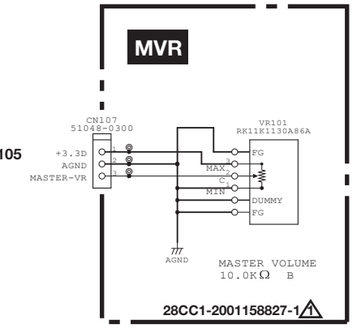

to DMLCD-CN301 <Page 1 K2>

to DMLCD-CN303 <Page 1 K4>

to PNL-T-CN106 <Page 3 D5>

28CC1-2001158827-1 ⚠1

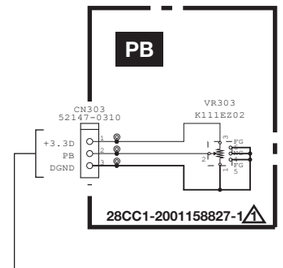
## MVR




to PNL-T-CN105 <Page 3 D6>

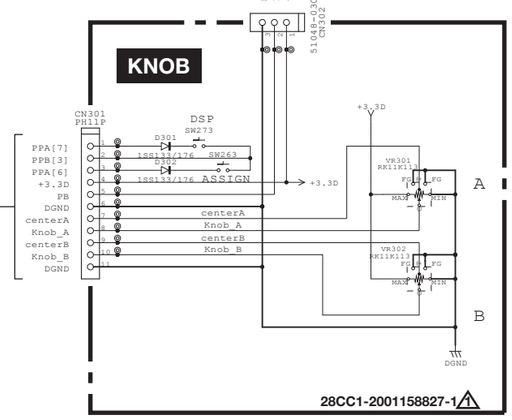
28CC1-2001158827-1 ⚠1

## PB




28CC1-2001158827-1 ⚠1

## KNOB




28CC1-2001158827-1 ⚠1

## PSR-E463 OVERALL CIRCUIT DIAGRAM (ENC,PNL-T,PNR-T,DC-IN)

< Page 3 >

### ENC

ROTARY ENCODER
EC351
REB161(9X5)PVB15FHINA1-2-24PCE

CN351
51048-0400

EN/A
DGND
EN/B
DGND

to DMLCD-CN304
<Page 1 K4>

28CC1-2001158826-1 /1

### PNL-T

PPB[0]: PORTABLE G SW240 D; DEMO SW250 B; TRANSPOSE SW260 B; MELODY SUP SW270 B
PPB[1]: LESSON SW241 B; REC SW251 B; METRONOME SW261 B; TEMPO/TAP SW271 B
PPB[2]: SPLIT SW242 D; DUAL SW252 D; HARMONY SW262 D; ARPEGGIO SW272 D

D204 1SS133/176 — PPA[4]
D205 1SS133/176 — PPA[5]
D206 1SS133/176 — PPA[6]
D207 1SS133/176 — PPA[7]

CN106
PH7P

to PNL-B-CN203
<Page 2 I5>

### PNR-T

PPB[4]: MDB SW234 B; 1 SW244 A; 2 SW254 A; 3 SW264 A
PPB[5]: FUNCTION SW235 B; 4 SW245 A; 5 SW255 A; 6 SW265 A
PPB[6]: CAT - SW236 B; 7 SW246 A; 8 SW256 A; 9 SW266 A
PPB[7]: CAT + SW237 B; TEN - SW247 A; 0 SW257 A; TEN + SW267 A

D215 1SS133/176 — PPA[3]
D225 1SS133/176 — PPA[4]
D235 1SS133/176 — PPA[5]
D256 1SS133/176 — PPA[6]

to PNR-B-CN252
<Page 2 D4>

51048-0800
CN151

28CC1-2001158826-1 /1

CN105
52147-0310

to MVR-CN107
<Page 2 K2>

CN104
51048-1000

to DMLCD-CN801
<Page 1 K6>

C106 0.01u(±)
C107 0.01u
C105 1000/25

D106 1SS133/176
D105 1SS133/176
/PSWI
PSWO
SW101
STANDBY SW

2SB1568 TR101
R101 1/4W 4.7K
R102 1/4W 3.3K
TR102 2SC5395
R103 1/4W 10K
R104 1/4W 4.7K

No Mount
C102 CHIP
C103 0.1u
C104 0.1u
L101 CMD8BB01
D103 LT2A02-E
D108 LT2A02-E
J102
J103
J104
FZ101 SCT-2A
D101 D102 10ESDA40-PAIB2

CN103
XH3P
BATTERY
1.5×6(1.2×6)

### DC-IN

JK101
DC-502-AG-PBT-2.0-B

52147-0610
CN102
51048-0500
CN101

PSR-E463:12V

28CC1-2001158826-1 /1

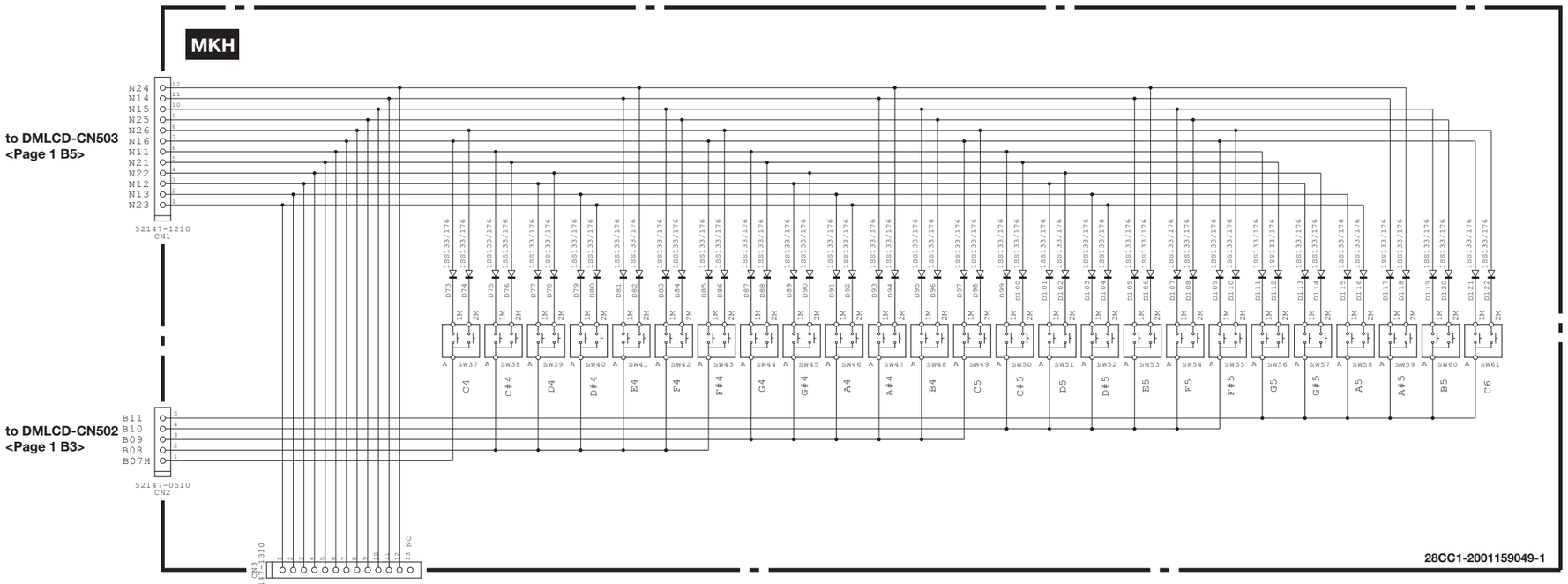
JC2.0
JC2.0

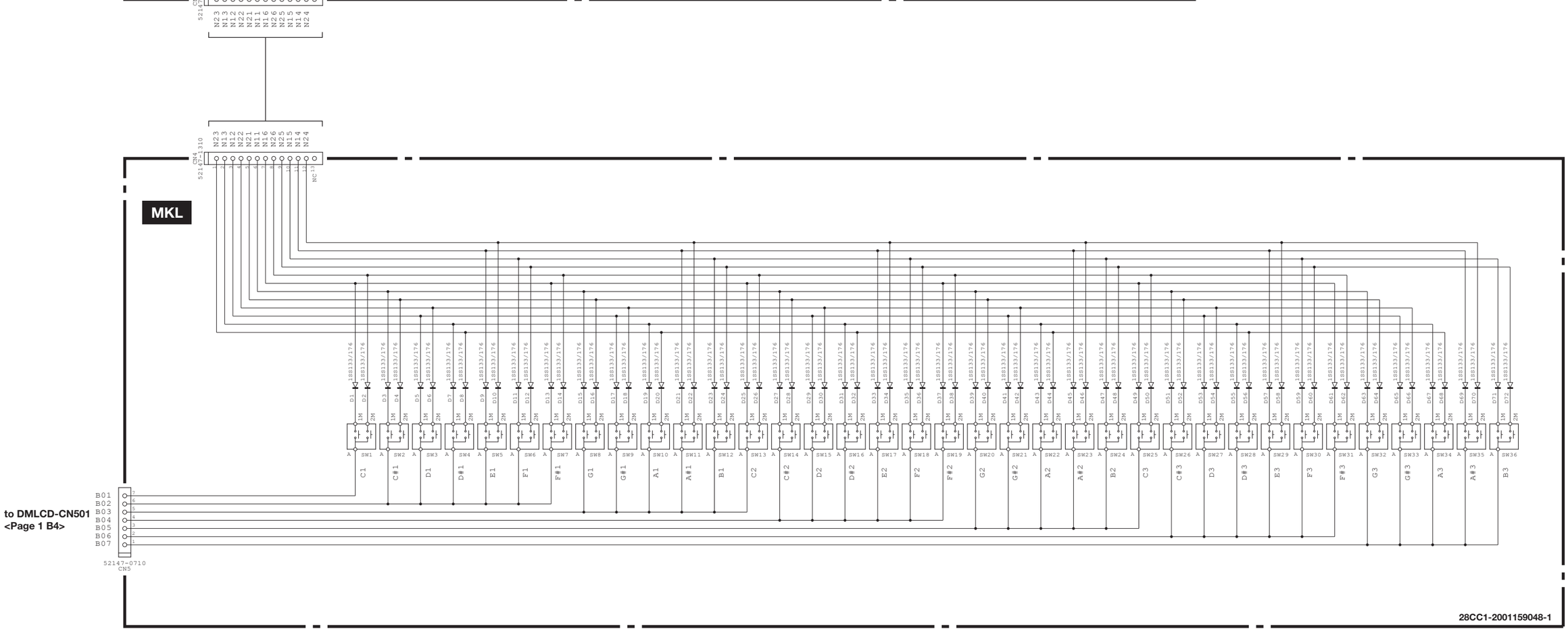
AGND
GND


PSR-E463

## PSR-E463 OVERALL CIRCUIT DIAGRAM (MKH,MKL)

< Page 4 >





PSR-E463